

Krys Dolega ~ Active Inference Insights 022 ~ Emperor's New Markov Blankets

Insights | Active Inference Insights 022 ~ Emperor's New Markov Blankets, Attention, Dan Dennett | 3:09:21

hello everyone and welcome back to the show today I'm extremely pleased to be speaking with Chris deLoraine Chris is a post-doctoral fellow working in the center for research in cognition and neurosciences in Brussels he focuses on computational models of cognitive functions mental representations and Consciousness as well as broader topics prominent in philosophy of computation philosophy and ethics of AI metacognition and philosophy of science Chris thank you ever so much for joining me um thanks for the invitation oh it's an absolute pleasure um people have been asking for a bit more of uh you know critical voices on active inference which uh and and you're the perfect person to go to but before we get there uh we were talking a little bit off air about Dan Dennett um Dan passed away unfortunately last week at the time of this recording I believe it was and he was your PhD supervisor so clearly a great deal of your work has been influenced by Dan um I wanted to talk about your 2015 paper um discussing the cartesian theater and it's sort of place or lack of you know it's it's it's sort of ought to lack of place in predictive processing or perhaps not but before we go there um I know you want to say a couple of things about Dan and it would be great just to discuss his sort of influence on you more more

broadly sure thank you so yeah so this is a really sad moment because that has I think it's easy to say defined in large in many ways defined the the the shape of philosophy of Mind in the 20th century up to today and it's a really great loss to to see him go um and I I guess I I mentioned to you when we were setting up this meeting that I would like to spend a little bit of time discussing him because uh I think that Dan is in some ways uh a kind of a philosopher who should be uh followed in the way that he approached philosophy of mind and uh I think that it's you know worth focusing on that that's that's one thing that I find really interesting and important about him and the other thing is that uh I had a personal relationship with him uh I did two extended research stays with him during my PhD and we've met at many occasions at different conferences uh and uh over the past weekend there were a lot of obituaries uh that that showed up about Dan and uh he clearly was a lot of things to many people and it's actually astonishing to see how many people he has influenced across multiple disciplines but I think it's also worth remembering uh that he was just an extremely kind uh nice guy he was just a he was just a joy to to be around um and so that's the second thing that I sometimes feel is missing from

contemporary philosophy so so the way he did philosophy is that he tended to focus on science right and he thought that there's no philosophy free science uh and similarly he thought that that philosophy of Mind should be informed by uh our best state of scientific knowledge and sometimes I feel philosophers forget that it's very easy to uh stay within the boundaries of our discipline and and not reach out uh and not try to to you know have a peak what's going outside of the boundaries of the Ivory Tower of philosophy of mind and I think it's really important to remember that that it that you know that if there's a legacy to Dan's work it's that philosophy can be informed by science and it's actually very productive and important for philosophers to engage with scientists and it's a kind of a two-way it's a two-way uh exchange of ideas where both sides can benefit productively um and then the second thing is that philosophers can be extremely serious about themselves and about their work and about really entrenched in their positions and and uh very competitive uh and uh yeah kind of hardheaded and I think that Dan uh although we cannot say that he had a small personality he was definitely uh you know a flamboyant personality as well but he he was really a person who was

very kind towards Junior Scholars he was
he would always find time to engage with
with uh younger researchers and
students he was very curious of what
other people might say about his work
even if it was

critical and uh he did not take himself
entirely seriously and I think this is
something that's also worth remembering
right so he he was happy to sometimes
concede that maybe he didn't think
something

through uh and uh he thought that first
philosophy first and foremost should be
a great intellectual adventure and not
some kind of

dogmatic holy war and I think that's
that's something worth remembering I

don't know if you would like to ask
anything else about Dan or well I um
yeah I mean firstly to Echo everything
you just said I mean I never met Dan um

I guess I'm I'm not currently at that
place in my career where that would have
been viable I mean um I only left school
what you know uh in

2017 but you know even studying
philosophy at school Dan's name was
pervasive right I mean interestingly
pervasive across sure philosophy of mind
which we studied but even qu you know
broader questions of epistemology um an
ontology Dan's name would come up in a
way that other philosophers wouldn't and
so he always seemed like a kind of

Grandfather of modern

philosophy in you know and so he's one
of for me one of the sort of icons who

I've never met and I've never SP spoken with but are incredibly influential even if as you say not I don't agree necessarily with everything I think Dan thinks or thought but yeah incredibly important um and you're absolutely right I mean just being on Twitter or X or whatever this weekend it was unbelievable I I mean maybe in some ways this is just because I I follow so many academics but just the timeline was absolutely replete with with uh obituaries and condolences and it was quite remarkable so clearly he touched a lot of people um both as a human being and also as an intellect yeah I'm curious what was he what was he like sort of during your PhD uh I mean I guess this is a personal question because I that's what I want to be embarking on you know this year but what was he sort of like as a as a guide and a mentor so he was extremely uh kind as I said before so so uh very encouraging he was eager to learn so something that I've heard from other people who have done post appointments with him or have been his PhD students in the past was that when someone came to him with an next with a expertise in a field that he wasn't very familiar with he was really eager to learn uh from them so so when I was working on my dissertation which was on predictive processing um and that was a new topic back then um he was very interested in learning about the models and And

discussing with me formal papers right about about different modeling algorithms for example which was super nice yeah um and one thing that was really special about him as a as an academic Mentor was that he was uh extremely well-rounded so I feel that today in Academia I mean it's not a product of the last 10 years but you know we we we've we have very narrow focuses we we we value specialization and then explored everything so a lot of people don't remember this about him but he was able to code for example he has done work with programming in the past uh he was very skilled in many domains and that would often come back because his curiosity and his abilities in in multiple areas would allow him to draw connections between different fields of work so so he was very quick in suggesting readings for example that you wouldn't get as standard recommendations from people working on a particular topic because they would be off they would often seem like they are very removed from what you're interested in but later on you would follow up and you would find out that there are actually connections between these ideas and that certain ideas surfaced in many different places in many different disciplines in Academia over time in different guises so that was extremely valuable uh and yeah he was just uh he was just fun to be around so he was

obviously very busy he was still quite generous with his time despite how many appointments he had and how many people were always kind of you know fighting for his attention and trying to to to get a bit of his time uh but one of the best pieces of advice I ever got from him was when I was visiting him and I was writing on writing a paper I was working on a paper and as I was halfway through this paper I don't know three quarters in someone else published a paper that was extremely similar to my paper to the point where I kind of thought okay I have to abandon this project because there's no way to publish this right now because it's just too

similar and I was pissed I was irritated I was kind of you know I can imagine I sunk a lot of time into that project and I went to him and I said yeah I'm I'm pissed this this just happened what am I supposed to do uh and he said to me something that I try to remember every time you know I feel scooped by someone with a paper or or

um someone is publishing like crazy on some topic that I'm interested in and I don't have I don't feel like I can catch up sometimes uh with the literature which has happened with predictive processing quite a bit my PhD uh because that literature exploded at the time but he told me this he said philosophers arguing about who came up with an idea first are like sailors wabbling over who

was the first one to notice that the
wind has
changed and I thought that's that's very
good advice
because the reason we all got into
philosophy wasn't probably because you
know we wanted to
be uh a superstar of philosophy it isn't
because we hopefully it isn't because we
have humongous Egos and we think that we
are right and everyone else is wrong but
it's just because we appreciated
thinking about certain
problems and it's in at the end of the
day it's a collaborative Endeavor and uh
you know science philosophy
uh building knowledge is is a project
where which in which we should engage
collaboratively and despite being a
giant that he was he was also then I
mean he was always very uh
uh in his writings and in in talking to
him he was always very much against the
idea of a singular genius so this this
idea which philosophy often suffers from
even today this myth of a you know
single author that can lift a whole
field it's a bit ironic that Dan was the
person who was against this idea given
how much influence he had over the field
but he and this is visible in several of
his books and writings he didn't like
this idea a singular genius he thought
that you know really certain
ideas uh start to kind of Brew in in in
in the right conditions and then there's
just someone who who who makes the
discovery or who catches wind of what's

happening but he always thought of himself as standing on the shoulders of giants like Quin uh and and and other people so yeah yeah that's a beautiful question qu it makes me think there's a there's one by KL young where he says people don't have ideas ideas have people yeah Yung was thinking about in a slightly different way than Dan was but it's um you know in in some psychoanalytic manner but the the sort of sentiment is the same yes and it's actually some it's extremely close to some of the ideas that he presented in from bacteria to B and back so so when he talks in the chapters where he talks about memes yeah it sometimes goes extremely close to to that line of thinking that uh we I was doing my first research stay when he was finishing that book and I had a privilege of of uh going through the drafts And discussing the drafts with him and Martin bowry who was a post at the time with him also a visiting post do and uh I remember that we s one evening and we had this conversation where we picked a passage from the book from the chapter about memes and we said Dan it's what are you sure this is this is what you want to say here because it starts to sound like humans are just you know material vehicles for memes and is this it sounds very extreme and I remember he was sitting there quietly for a second thinking and he said you know what maybe

you're right maybe my Pros got the
better of me
so um yeah so so to say right people
always uh praised him for his Pros right
um famously are kind of
very well praic is a derogatory word but
I mean it here in a in a sort of
praiseworthy way that you some
philosophy is very snooze worthy let's
be honest uh but you know the things
that I read of Dan are certainly not and
that itself is quite remarkable um you
know the ideas aside of course which are
very potent and influential to be able
to write about something otherwise
deeply complex and otherwise for you
know a certain audience extremely boring
and to make that come alive is an
incredible skill in of
itself I yeah go ahead I just wanted to
say on that point that I completely
agree with you and uh he definitely had
his own voice as a writer which was
remarkably close to the way he would
speak in
conversations uh but what's really
interesting is that that way of writing
that he introduced into
philosophy uh it was it it it gained
praise over time I think late in
especially in his later years but
initially it was a bit of a problem for
him so when he wrote his first book
content and Consciousness which I
believe came out in 1969
that book got some negative reviews
because of the way in which it was
written and it wasn't a

fully actualized denian narrative yet uh
but you could still get some you know of
his uh his kind of Massachusetts
sensibilities in there and and what was
really interesting is that uh some
people really
uh recoiled from that style of writing
at the
time well I think um yes I I sense
there's been a change in that right I
mean the the underlying soci
sociological reasons for why that's the
case whether it is just uh people
copying someone like Dan is an open
question but I do think that um well
Duty might be a strong word but we have
a we we do have an obligation in some
sense to make our works leg
right uh even if it's not for the L
audience even if it's being published in
philosophy journals we can't be talking
into you know this the N of one um you
know V chamber that is just our own mind
and Dan again of the stuff I read from
and also his contemporaries uh people
like Ned L and and David Chalmers and
then some of my favorites you know Evan
Thompson who I know was a postto with
Dan it's it's fun to read um and it
makes people like me want to do
philosophy and again this is no shot at
anyone who was working 120 years ago
whose Pros might make you want to fall
asleep but um I yeah I guess there's a
broader broader thing there that um we
do have a we do have a duty to transmit
our ideas in a legible intelligible
Manner and Dan epitomize that definitely

and he was also I mean he was also a he
had the style of his own which was very
entertaining which I think is another
thing yeah a lot of a lot of academics
could learn from him uh he's not the
only example of that
um but but of of of you know of being
able to write in an entertaining way
while also transmitting very complex
ideas but
uh yeah I like this idea of an author
having their own style you know um I
think Carl is one of them I mean it's
fitting that we on this podcast right I
mean um I've been accused at certain
points in my writing of writing in
frisan E I love I love review's comments
where they mention Frist e because I
can't help you know having read so much
of frist's work of Carl's work I can't
help but use the word
evince yeah that's definitely that's I
feel like it's a bit of a dog whistle
you know it's a if if someone uses the
word evince in their paper especially
when you're reviewing a paper you can
see who they've been reading uh a lot so
so I feel like evince is this kind of
Mark of of someone who has read a lot of
friston if EV appears in your own
writings it might be because you've been
reading a lot of friston and moving away
from Dan a little bit and and towards
what you wanted to talk about in terms
of papers I reread some of my older
papers for this meeting because you know
some of them are seven years old and I
don't remember what I wrote there

uh I was yeah definitely it's a journey for me as a non-native speaker to develop some kind of style and and I I read some of the sentences I wrote in the past and I wouldn't put those ideas in that way today but you know well I've got to say that I didn't have that impression um the one I read I think it was yesterday was the The Curtain Call for the cartisian theater which was 2015 which feels I mean like yeah I was doing my GCS for people in England you know I was doing my sort of I was 16 or whatever but um I didn't feel that way at all and and clearly there's a lot of denti and influence there in in that paper but what I like about it I was going to say this in reference to Dan is you know you mentioned his kind of openness his curiosity his epistemic humility fundamentally you know I mean we don't know with with a lot of things in life we really don't know and that's the beauty I mean we would all be out of jobs if we knew and life itself you know there would be no information transfer arguably there would be no life at least worth living um and I felt that in that paper you very much evinced if you pardon me the word a great deal of epistemic humility which I really liked I mean it you know you're critiquing the Frisian in this case the Frisian hsian model of Hobson model of um their their sort of invocation of the cartisian theater and what Dan called cartisian materialism when it comes to I believe you call it action orientated predictive

processing but I guess we call it active inference now

yes but you know you didn't say but this is the this is the fundamental way in

which Consciousness appears or this is the way the the representations work so

I enjoyed that I mean for people who

haven't read that paper what and again

it comes back to Dan what is cartesian

materialism um what kind of role does it

have to play in terms of predictive

processing and then what's the what's

the problem with it you know okay uh

nice so we're really starting at the

beginning so that's a paper uh me and my

good friend Joe with whom I studied in

edinburgh before starting my PhD uh we

studied in edinburgh uh where we read a

draft of Andy Clark's surfing

uncertainty and one of our Master

seminars wow and that's how we got into

processing um so so that's a paper we

wrote early on and it was a kind of a

knee-jerk reaction to a paper by Carl

friston and uh Hopson yeah H in which

they invoked the notion of cartesian

theater so cartesian

materialism is this term that was coined

by Dan dennet to describe a certain view

of Consciousness which he thought was

very a kind of a residue from the from

the ideas of daycard and you know the

idea that there's a certain point in the

brain where information needs to reach

and then it somehow needs to be rendered

in the medium of Consciousness so for

for decart that was of course the pineal

gland which was this point of

interaction between a mechanistic
mind uh and the soul and there was a
there was a certain in his view of the
mind and his of the brain also
uh there were these uh humors right
there were these uh different liquids in
the brain that would create pressures on
the on the pineal gland and depending on
which part of the pineal gland they
would put pressure on there would be a
certain

experience that would uh appear in in in
the soul in in

Consciousness uh so so what denet
criticized about this view of

Consciousness is that it requires a
certain type of double trans
transduction he called it so it's not
only that there's sense it's not only
that there are forces impinging on our
sensory periphery from the environment
which are then transduced into the
medium of neuronal spikes and you know
um hormone secretions and so on all the
things that are happening
mechanistically in the brain but then it
seems to imply that there's a certain
point in which in the processing of
information where

the medium information needs to be
transduced again from the medium of
neuronal spikes and all the
mechanistic um happenings in the brain
into the medium of Consciousness so he
sometimes referred this this this the
product or the this this
this substance of the second transaction
he called it figment right with with by

analogy with pigment so pigment is this thing that needs to be put on to your you know onto your representations of the world in order to imbue them with qualitative properties because presumably as most dualists and non-reductive physicalists would tell us there are phenomenal properties and there are dispositional or functional properties and these are two different types of properties right so all anti-reductionists about Consciousness will claim that phenomenal properties are not reducible to functional properties and then said well if that's the case then it seems to be then we end up with this picture of the Mind where information hits the retina let's say in the case of vision it's transduced into the medium of neuronal spikes that information is those neuronal spikes are propagated into through the cortical hierarchies and then we have to ask where is this point in the brain where we put on the pigment onto the onto the visual information right so so it seems like we need to apply the mental paint as block would put it again even though I would like to say that here as I'm talking about mental paint that's a slight abuse of the term as block uses it but the point is that you need to you seem to require adding something into the workings of the brain in order to render information consciously information conscious and that's

something that he really wanted to reject so his point uh wasn't that there's a his point wasn't that there's there cannot be some aspect of the brain which is privileged in terms of bringing about conscious experience H it's rather that there's nothing extra in ter metaphysically extra or ontolog ically extra added into the information when it becomes conscious right it's just that you have you process information unconsciously and then it can become conscious at any particular time uh given that it was processed enough through the cical hierarchy and so on right so so there's no mysterious properties that need to be applied again or added back into the into the information that is being processed by the physical system so he very much so the name cartisian Mater materialism is supposed to suggest that there's a there there's a materialist picture of the brain but there's this much like in the case of the pineal gland in in the case of thearts dualism in cartisian materialism there's a particular location in the brain that information needs to reach in order to become conscious and this and this particular Center in the brain is this one area that serves as the cartisian theater kind of as a as a place where which serves as the screen of Consciousness right and that's something

that he wanted to reject because he believed that there's no one particular place in the brain where information needs to be processed in order for it to be conscious it's rather that there's a collection of different functions that need to be in place simultaneously trading information in order for information to be conscious right and that's also why in many disorders of Consciousness such as I don't know visual form agnosia uh or blindsight we see that conscious experience undergoes a kind of graceful degradation where people lose certain aspects of conscious experience uh but that doesn't mean that they are rendered somehow entirely unconscious or something like that right it's it's rather that you can knock out different modules if you want to talk about about that in this those ways are different systems out but as long as there's some kind of wider Comm of information going on throughout the brain Consciousness is still in place so I feel like the rebuttal to this would be the argument that okay I guess the cartesian theater in some ways has multiple reference in the sense that one thinks of as you say the screen of Consciousness I mean the word theater brings that to mind very vividly uh and one thinks also of the homunculus who's watching the the screen and this sort of little man who's watching and and that's obviously got a number of problems most strikingly the infinite

regress of the homunculus but then one
but then there's also this separate more
mechanistic point about the locus of
Consciousness and this sort of singular
physical point I guess the rebuttal to
that would be well even if you adopt a
more distributed idea of the brain as a
network and you're talking more in terms
of functional connections there still
seems to be either you know either we're
saying well there is no
consciousness well let's disagree with
that just for the meantime but let's say
there is consciousness well we don't
want to say that necessarily the sound
waves or the visual information itself
is conscious so there does seem to be a
transduction at some point from matter
into phenomenal space now that sounds
like to me that there needs to be a
moment at which just by necessity that
information enters into you know in
metaphorical use that phenomenal space
now it might not be the case that that's
the same for every experience it might
not be the same that it's always it
might not be that it's the case it's the
same in every place for you know for
every experience but it does seem to be
this case that even in a diverse range
of networks there is a moment at which
the lights come on so to speak would Dan
or yourself have a problem with that
weaker form of the cartesian materialism
or cartesian theater
um all right so that's a good question
that's you're pinpointing exactly the
the the main issue so

so

um

Dan I think would have a problem with that because especially so so he explicitly wanted

to uh leave behind this metaphor of a screen mhm or this metaphor of uh you know know some kind of special properties in in Consciousness so a lot of people attack him these days for being an eliminativist about

Consciousness and he was very adamant that he's not in that camp precisely because he would always claim you know there is something it is like to be Dan dennet and it's different from what it's like to be me and it's different from what it's like to be you so there is a subjective perspective on the world that you have that is that is different from mine and that was different from his uh and uh so he didn't want to deny that but he wanted to change the way we speak about these things so so this is why he together with Kei Frankish they coined the term

illusionism because they wanted to to stop talking about phenomenal properties as things that we have to uh find a place for in our ontology now

uh yeah so that's that's one thing and I think that he would also have a bit of a problem with that because uh of his multiple drafts model so so his idea was that you know he was very much informed by uh bars and thean and the GL the work on global workspace so this idea that

there's

this there is this there are these

longterm long range projections sorry

long range projections in the braak

which connect different areas and

there's a kind of exchange of

information going on but on his view of

Consciousness there isn't one particular

system that is the Consciousness system

right so so it's it's likely that I

think he would agree with the claim that

a weak claim that there is a number of

fun cognitive functions that need to be

in place in order to for something to

have Consciousness like ours or

sufficiently similar to

hours um but I think he would reject

the uh the claim that any one of these

systems is somehow privileged over the

others that's that's one thing and then

the multiple drafts idea is that there

are multiple competing streams of

information in all of these systems that

are necessary for

Consciousness and which stream

you're conscious of which stream of uh

you're experiencing is kind of up

to one the the the how much Cloud it has

in the system so how strong how much

control it has over over the rest of the

of the cognitive economy what he kind of

as Fame in the brain and then the other

thing is the kind of probes that you

encounter so so kind of uh either

external um external situations which

which or stimulate that that that

require solicit a certain response from

you or internal probes like you know if

you're if you're sitting there and
you're really thinking about your uh ex
or if you're really thinking about the
EXP focusing trying to be mindful about
the experience of eating an apple that's
also a way of privileging certain
streams of sensory information or or
memory traces over other streams of
information in the in the brain
um and so okay so that's I think what he
would want to say probably about this
now for me I guess I'm
less uh I'm less
uh convinced of of you know I I I like
the multiple drafts model but I'm not
completely on
board uh because a lot of the work in
science of Consciousness these days
focuses on uh on higher order systems on
this division between lower order
systems and higher or lower order
processing and higher order processing
metacognition um and I think
that I'm leaning more towards some kind
of higher order theory of Mind these
days and I do think that there is a
there might be a certain
privileged set of processes or systems
that are are are necessary and in some
sense they
might that they're not exactly I
wouldn't want to slide back into
cartesian materialism by saying like oh
information needs to go there in order
to be conscious but they do play an
important functional role um and I think
that because I I I subscribe to some
version of the higher order View I think

that means that I have to be a little
bit weaker on the cartesian
materialist critique of these
days because there is a certain I do
separate the system into two levels and
where one level seems to be
privileged over the other in terms of uh
the role it plays in bringing about
Consciousness okay cool we can go there
that's that's that's an interesting
development because um I'm so I'm aware
of these higher order theories of
Consciousness with respect to
self-awareness so as you know there are
these
so-called uh pre-reflective states of um
self-awareness or self a subject which
you can sort of deflate just a
subjectivity let's say and the whole the
whole principle of these you know
phenomenologically and ontologically is
that no one there is nothing that is
aware of them they offer the uh texture
the background to uh the objects of
Consciousness but they never areed into
an intentional object obviously his work
goes back to Spinoza and so on now uh
this kind of poses an issue for the
higher order theories of Consciousness
because for them these states or or or
these experiences which in turn are not
reflected upon they well it seems a bit
problematic because they propose that
actually there are higher order states
which take these conscious uh these
pre-reflective states as their object so
there is no such thing as the pre-reflective
or non-reflective self-awareness rather

it's sort of hidden hidden uh these
higher order uh intentional as are
hidden from us and then the rebuttal
comes and says well how you know how are
these not conscious right like surely I
should be aware of these higher order
states which have as their object my
pre-reflective states of self-awareness
and the higher order Consciousness
theorist come along and say Well they're
not conscious uh and they intend the
pre-re states of self awareness without
themselves being conscious and then you
have just the ontological question well
how does that work and so I guess the
question would be we seem to have these
it doesn't have to be self-awareness we
seem to have these uh phenomenal I guess
the word coming to mind again is
textures right I mean what people
traditionally would have called qual or
Quail which is sort of um what you know
seems to constitute the what it is
likeness of Consciousness so the redness
of red or the APPL of eating an apple
and so on and
um these seem to be constitutive of my
experience even if I don't necessarily
have to rify them and that's the same
with sort of the my own subjectivity or
my own intentionality how do you as a
kind of leaning towards a higher order
Theory Of Consciousness how do you deal
with these otherwise seemingly
irreducible ineffable intrinsic features
which at least on the surface seems to
seem to
Bell um becoming an intentional object a

sort of overt reflective

object uh all right wow uh that I'm no

no no no no that's that's that's a good

question um so I must say that I'm not

that familiar with the literature on the

pre-reflective uh self-consciousness I'm

I'm I'm getting more into it because I

was recently interested my my recent

work is more interested in affect and

effective experiences so that's what I

work on in Brussels now and some people

postulate that for example in order to

have emotional experiences you need to

have

a an explicit involvement of the self

yeah so jolu for example is a higher

order theorist who claims that in order

to undergo an emotional experience you

not only need to activate what he calls

an emotional

schema uh which

conceptualizes your the the sub uh

subcortical processing in the amigaland

and so on but also you need to have an

active self schema so you kind of his

big big catchphrases no self no emotion

so unless you you are aware that it's to

you that something is scary then you

just don't feel fear right previously he

call like the

smaller uh claim was no self no fear so

you can only fear something if you

realize that is dangerous to you or or

you you fear um that thing that object

so so I'm not yet that familiar with

that literature and I know also that

even if we leave the self aside there's

this whole literature on minimal

phenomenal experiences which is something that Thomas Metzinger is now very interested in. Um, precisely because in deep meditative states you can let go of the self-right and you can presumably have these pure phenomenal experiences, uh, which are supposedly, uh, relaxing this idea that there's a self which is the anchor for the point of view on the world. Um, and I think that you're right that both, uh, minimal pre-reflective, uh, experiences, uh, this and minimal phenomenal states, they do put pressure on higher order theorists, and so I think that there's two ways in which we can respond. So, one way is I think the way that would be taken by someone like David Rosenthal. So, David Rosenthal talks about qualitative spaces, and he very much wants to, uh, want to uh, reclaim the word qualitative property from the hands of strong phenomenal realists and say, look, there's nothing mysterious about qualitative properties. As such qualitative properties as such are not, uh, some kind of mysterious properties, they're just we know about qualitative properties we talk about qualitative properties in science, and we can think of the brain as representing qualitative properties that's not the problem, regular scientifically.

understood qualitative properties are not problematic on their own in in in a deeper metaphysical way unless you conflate them with qualia which are supposed to be not only qualitative but also you know they're supposed to be ineffable they're supposed to be private and so on intrinsic and so on and so forth so what he's trying trying to do is he's trying to separate the notion of qualitative property from the notion of intrinsic property and once you do that you can actually claim that on the first level you will have representations of qualitative properties and then you need this kind of higher order process that targets or picks out or re- represents depending who you are these lower order States in order to have qualitative experiences and that's that's one way you can go about it I agree that this might not be satisfying to you because the problem then is well okay so what about these states where we seemly undergo ego uh dissolution and and where the self seemingly doesn't play any role it seems to me that in those cases traditional higher order theorists will have to claim that well you're kind of in a under an illusion that there's no that there's no self perhaps The Narrative self Falls away but there might be this kind of minimal notion of a self that's still in place for example presumably if you're in a deep meditative State uh you're still

deploying your attention in some way or
controlling your attention in some way
there's some kind of process of
attentional monitoring
happening and the system which is
controlling the the the allocation or
the flow of attention if you want to
call it that that's the system which is
actually responsible for for this
minimal self that's where the minimal
self resides and and Michael graciano as
you probably know has built a whole view
of Consciousness around the idea that
the control model for attention is the
the the the representation that we
introspect when we're thinking when
we're experiencing things and then the
second way so that's one way to do it
the other way would be to say well how
many higher order levels are there right
so you can always this is a you
know it's as much of an objection as a
as an argument in favor right so you can
always claim that there's an extra level
of representation and it's not that
things have to be represented on the
second level there's a higher level and
so for example in graciano's model to go
back to that as I understand his picture
these days we have have some first order
perceptual or or other
processes uh then we have the
attentional model which we represents
these systems in a kind of uh way where
they are
distorted uh and then we when we
introspect there's actually a higher
order model yet that is picking out the

contents of the attentional control model uh at least that's one way to understand his claim and so in that picture we would have this kind of of three layers uh of of the higher order model right so it could then we would we could claim for example that when we're having these pure minimal experiences that the higher the higher level that this introspective level which which renders things as for me experiences is dropping out but the control model again is still there in place and so what we're experiencing is just these representations in the control model but there's no explicit level of introspection so the self is kind of sidelined I'm kind of spitballing here right now no so so was I to be fair because as I said I'm not uh first of all I don't have a complete Theory Of

Consciousness and I would be I hope the last person to claim that maybe one day uh you know I will feel more confident in claiming some piece of the cake for me for myself sure um but but and I as I said I'm interested in these minimal States uh of either pre-reflective self-awareness or minimal states of phenomenal experience but I don't know enough about them right now to you know give you a very um strong position about them yeah I mean to be thought I was somewhat spitballing as well I am not an expert in higher all the theories of Consciousness I mean I'm aware of David

Rose

and others who have proposed things akin
to a high World theories of
Consciousness right I don't really
believe in sort of Monolithic these kind
of categories right everything is sort
of everyone's got their own
idiosyncratic way of viewing these
things um I I came across it I I'm very
I am interested in pre-reflective self-a
Awareness kind of phenomenologically
more than anything I mean the
ontological metaphysical ground of it is
an interesting question I I mean just to
State my claim at least on something
like subjectivity I think there's a it's
absolutely viable to take a kind of more
San or moonan position that it's
actually just imminent it's intrinsic in
all experience even in minimal
phenomenal experiences there's still the
for
meanus and then I mean this feeds into
my kind of skepticism over higher order
theories because even in this notion of
a kind of higher order representation
there is already inbuilt into that the
for meus that is supposed to be being
represented which seems to be kind of
begging the question or you know putting
the cart before the horse
and I would again I mean this idea that
it's just intrinsic to experience and
then in some ways you're saying well
experience is intrinsic fundamentally
and there is just this
irreducibility of course that comes with
its own problems right I mean I mean how

first and foremost I mean unless you're just a Jew list out the you know out the can like so so no these are interesting metaphysical questions I guess um it was probably it was probably a bit misguided of me to bring in the notion of qualia because well I guess this is an interesting question when I was thinking about it I was thinking about the way it sort of pre-reflectively uh structures our experience such that I I mean this again I don't really know if this is true do I need to reflect on the taste of coffee to experience the taste of coffee in the same sense do I need to reflect on my I don't well seemingly phenomenologically I don't need to reflect on myself as an object to experience myself as a subject um and so that's kind of why I brought in qualia but I think probably the higher order theories a better chance of dealing with qualia right it's just a representat of some represent you know representing scheme um yes I mean these are Dee rooted philosophical problems that we're not going to be able to solve on this podcast but I think an interesting question here is um bringing it into a basian account so you brought you mentioned s of Fame in the brain there uh another one of your papers s Fame in the basian brain yeah the question I think that comes up just to mind is if we took denit multiple draft models seriously and it seems It's well in some it seems incredibly convergent with

predictive processing and active inference in the sense that we're running multiple hypotheses of the underlying cause of our sensory data the thing is is that we in principle you're never reaching the kind of infinite peak of a gaussian you're never coming up to a you know no hypothesis in principle statistically should occupy a one from 100% certainty because this is just statistical distributions right yeah if we're taking a very Bare Bones approach to basian the basian brain hypothesis so I mean I've been reading this in terms of people like Andy Clark and Kate Nave have spoken about this kind of seeming in the fact that perceptual experience seems determinate or be it seemingly if we take the basian brain hypothesis seriously underpinned by indetermined probabilistic processes so I'm curious about how one can skirt this problem I mean Kate has her own solution which is that Consciousness in some sense is indeterminate although not in the way that people naturally think of indeterminate she thinks as a sort of singular reality nonetheless um people like Jakob Hye will talk about distrusting the hypothesis and so there's this constant temporal flow I'm curious about how you can get how you think we can get from a predicted processing picture of multiple hypotheses being entertained to a singular seemingly singular coherent global of conscious

perception yeah that's a that's again a very good question um so I think I would start this by saying that I don't so so I'm very much interested in the basent brain hypothesis as a from a kind of anti-realist stance right so my stance towards towards these models is kind of um disengaged in a certain sense ontologically so I suspend my belief in them as actual descriptions of what's happening in our heads yeah and I think that uh I take this removed position of all models are wrong some models are useful some models are interesting uh and I think that actually probabilistic models have a lot going for them but it's just one of the possible ways of capturing what's going on in our heads uh and so we always have to so if you approach them from this this point of view then then you know first of all it relaxes a little bit your commitment to finding the one true solution so so you can say oh we could model this in this way or we could model it in that way we we make some modeling choices that in the end determine how we're going to think about it or how we're going to model the behavior of the system uh and in some ways they are useful but in some ways they can also be misleading right so um so that's that's that's the kind of to preface my my answer here so that's the first thing uh following ing from that one of the big decisions that you have to make when you're deciding

how to model the Bayesian brain. If you assume that the brain is doing something like Bayesian inference, then of course the question is how are you going to model the process by which it's doing it. So is it we know that it's highly unlikely that it's running exact Variational inference, uh, so it needs to use some kind of method of approximation. So Carl and people in his camp have settled on variational Bayesian inference as the most likely solution here, but there are other options as well. Right, so you could think about sampling, for example, which is another way. Yeah, and if you had a model based on sampling, you wouldn't actually have to be like that the problem of determinacy is gone in many ways because if you think that you're you know sampling from some kind of distribution, then you're and then you're experiencing a sample, then you're experiencing a sample, it's a determinate point and a kind of you the information is determinate, right, even though it comes from a probabilistic distribution. So that's one way you could get around this. Um, the other thing that I wanted to point out is that in uh, Jacob's picture of how things work, so Jacob talks about probability density functions of course, um, and uh, he talks about some kind of uh, as you said, uncertainty about you know the exact uh, hypothesis, right, that you're entertaining because you're never getting a kind of a pointwise

estimate you're always getting some kind of

interval um and that's that's also a way to go get out of it another way to get out of it is to for example bring a bring into the picture some kind of uh dual coding hypothesis so you could think that on certain levels of the predictive hierarchy things are being coded in continuous spaces and in others they are discretized right so some levels of the hierarchy are operating on uh on probability density functions and there are parts of the hierarchy that are operating on probability Mass functions okay where you can actually pick out one hypothesis as the as the currently winning hypothesis so the way you the reason why you might think that's the case is because of how Concepts have been typically handled in philosophy right so concepts are supposed to be these kind of distinct categories at least

traditionally nowadays A lot has changed in the way we think about Concepts but you could think that as you're bringing something into a conceptual frame that conceptual frame is somehow constraining your experience so you could think that well when I'm perceiving colors of course the space of my color perception is continuous but then as I'm recognizing colors as I'm applying certain Concepts to them I'm discretizing my experience in certain ways uh the problem of course with that is that there seems to be a lot more

variety in our perceptual experience than our conceptual apparatus this allows for so we both agree on what red is but we could find multiple examples of objects that we would categorize as you know we would disagree about whether they should be categorized as red or not right and there seems to be a lot of perceptual experiences that fall under the concept of redness for both of us and we would agree that there is variability in the in what we call in in the experiences we would uh uh label with the term red so so that's then a problem

and I guess another option sorry I'm kind of lay layering you know different different ways you could think about it and finally there's a there's another option which is

that uh even though you have multiple distributions that are kind of indeterminate uh or or have these kind of intervals where where you know you're modeling things with probability mass functions probability density functions so you're you're just picking out intervals you might think that if you have some process higher order process for example that takes information from multiple processing stages then there's a possibility that all of this is somehow collapsed into each other in a way where you do get actually very narrowly defined uh hypothesis right and again you might think that this would happen in some kind of discrete

medium uh that doesn't have to be
connected to uh Concepts
necessarily uh so so so I think there's
plenty of ways in which you can you can
get around this problem for me
personally I think that uh nowadays I'm
actually kind of removing myself from
this this debate and I just think that
well there those are multiple ways in
which you could
model conscious experience and then and
build models of perception for example
and and and categorization of perceptual
experiences and so on and what's
interesting is really whether these
models can inform new predictions or new
hypothesis that can be tested
empirically
right yes it's uh we will come I I I we
will come to the scientific realism
versus instrumentalism thing because I'm
sure people want to hear your your take
on that
um in Greater depth it's I'm really
sorry can I can I just add one more
thing so so another reason why you just
just to kind of again just talk about
this you know why would there be one
hypothesis that's winning uh one reason
for that is nicely given by yakob Hui
and uh franchises Mari in their paper I
don't 2020 I want to say I don't
remember when it came out yeah I'll have
a uh but y but they kind
of they they subscribe to kind of Jesse
Prince's view of mid levels of the
hierarchy relevant for Consciousness but
they have actually a very good answer to

why there might be your perceptual experience and or experience in general is not kind of indeterminate but is kind of rather fix fixed in a certain uh temporal window right because what's also relevant here is that that when we're thinking about experience we often tend to leave out time but time is a really really important aspect of our perceptual experience and our experience in general Consciousness in general we know that there is a certain integration window where you know stimuli that are presented in the same time will be integrated into one percept for example so the reason they give for why there might be a certain uh deterministic quality to our personal experiences and experience in general is that actions are actually not indeterminate right yeah every action you have to take in your environment is a kind of a determinate situation in which you're putting your body and the way you're changing your environment by acting on it you're moving things around and things kind of can either be here or there I can lift my arm and I can lower my arm right and I'm in between two uh states there might be some gradation between these two states but if I do one thing I can every state or every action I take or every state I put my body into that limits my possibilities constrains them in a certain way and my body can be only in one state at a time so when I'm performing actions when I'm planning actions uh I need to kind of

experience the world in a way where I don't experience myself doing five things I'm doing one thing and that's actually you know perhaps there were organisms further on if we think that the ban brain hypothesis is true perhaps there were organisms that had a more indeterministic experiences of the world and presumably they would have died out because determinacy yeah yeah exactly yes yeah that's ABS that makes perfect sense I mean again I'm I point people in the direction of Kate nave's work as well I mean she was just on and and she sort of piggybacks off yakob and um Mar's work on this uh sort of in what do they call it sort of intermediate processing right like these are all and again and perfectly chimes with things like affordances um and the fact that perception is enslaved by action to some degree I guess uh well multiple things that come to mind I mean one is it seems to me like a lot of this is uh comes down to the question of attention because we we make the claim that the world is determinate the perceptual world is determinant but I would propose and again this is riding off what Kate thinks that the world is determinant in the very sort of focalized purview of my determinate world now if I have if I'm looking at my periphery it is not determinate at all and I think if to also build onto this action-based picture it's because I can only really I only really Act on what I pay attention

to so do you think it's viable to say
that you know you're talking about this
higher order process which collapses the
um probabilistic densities into a kind
of you know we can think of it doing
kind of a direct Delta function it just
like collapses it into a single
value is how much of that would be that
that is actually just ATT tension right
that higher order process is
attentionally homing in on a singular
you know derives a singular per
and then we can say well that's because
attention is in many ways the slave of
action I think that's very plausible so
this is more less The View that we
defended with Joe in the fame in the
brain paper right so so we kind of put
attention in the in the Forefront of
Consciousness there uh this is also a
view s i I defended a similar view in my
dissertation uh and I do think that
attention is extremely important for for
Consciousness and know yakob actually
also thinks that right so he points out
that on predictive or or the active
inference stories about uh the
brain uh Consciousness cannot be
entirely separated from attention
because attention is precision
optimization right and precision
optimization or expected Precision
optimization happens on multiple stages
in the system happens on multiple stages
of the processing stages of the system
simultaneously MH so so there's I really
there's never a time where the
attentional mechanism is completely off

so to speak in the brain there's always
some kind of biasing attentional biasing
in terms of uh expected Precision way
weighing of of uh error prediction
errors happening somewhere in the system
so attention seems to be necessary for
Consciousness at least on on the on the
active inference predictive processing
story uh so yeah I I find Kate's
suggestion
uh very palatable how much does this um
feed into broader questions of
agency um teleology if you want to go
all the way there and and and relevance
I mean I'm interested in theories of
relevance or relevance realization John
BVI this kind of idea and the idea
attention in many ways is the kind of
driver or or the product of a organism
which needs to uh you know inact
relevance realization it needs to be
able to disambiguate that which is
relevant to it and that which is
irrelevant to it and that I think I I've
made this argument I wrote this paper
it's um currently in review about how I
think that attention is ultimately the
slave of preferences and I think you
could say attention is and that means
that attention is ultimately slave of
relevance because you know I it's not
random what I pay attention to obviously
and also the very fact that I can pay
attention is is itself a prior an act of
prioritization it takes a kind of
flattened world and creates peaks of
salience yeah so how much do you think
that we can build in values

relevance meaning and so on into this kind of more strip back mechanistic perceptual picture that's a really nice question and if you don't mind I would like to read that draft if you don't mind to because this is something that I'm working on now in Brussel with Axel Claremont and actually Axel has done some research years ago on attention and salience in addicts so he has shown he and other people that his Co authors they've shown that there is so so attention is driven by preferences as you said in the sense that addicts do pay more attention to things that are triggering their addiction right so you know if you're a gambling addict and I show you pictures of uh poker chips or cards those are much more Salient to you than kind of neutral stimuli right right right right uh so so so that's the the first point so so I think you're on to something there and I I just want to also say that yeah that's proba like this is this is um evidence coming from humans but if we go to a much more basic level I think that attention is doing basically prioritizing certain information streams over other information streams yeah um and what that means is just as you said organisms that have attention are probably organisms that live in envir en Ms complex enough where they have to do some kind of motivational tradeoffs or they have to tradeoff acting on one piece of information against another

piece of information right so there
there there seems to be attending to one
stimulus against another stimulus so you
have to
prioritize what you're going to do and
so if attention is in the business of
prioritizing processing of or trading
off processing of certain things over
other things then that's exactly very
similar to motivational tradeoffs right
where you're you're responding to a
certain stimulus over another
stimulus uh because you know even though
one stimulus for example will yield you
rewards immediately another stimulus
will yield you greater rewards in in a
in a window of remove time so so you
have and and these two things seem to be
quite connected in a sense that if you
want to apply yourself to you know
pursue a stimulus that will yield
greater rewards that are removed in time
it seems that it would be useful if you
had a some kind of a very riment perhaps
system of prioritizing information about
stimulus and kind of making you direct
yourself away from the from the
immediately rewarding stimulus right
yeah and I think these things are
extremely connected and in fact if we
look at the literature on addiction and
evidence that comes from rodents so
so uh work of people like Kent parage on
mice the main driver of what they call
incentive salience so so what has been
labeled in the literature as the wanting
signals so signals which which uh code
prefer the preference for stimuli um or

the drive for certain stimuli the these
uh these signals have been linked to
dopamine which has also been linked to
to attentional control at least in
humans and and other mammals so
Precision right exactly exactly so I
think that there's a there's there's a
lot
of uh there's a very strong relationship
here uh between these between attention
and some kind of uh I'm not sure I I
always worry about the word preference
because I worry that people will
misunderstand yeah what we mean mean by
that because when scientists work
scientists working on reinforcement
learning or you know rodents or or uh
some kind of simple animals they talk
about preference it's clear that they're
talking about
preferences uh but they don't talk about
some kind of conscious yeah intentional
States uh the way we think of them in
everyday life and I sometimes worry that
especially in philosophy of Mind where
people people often come from this
armchair position where they think first
about how things seem to them uh from
the first person perspective for example
and then they they they look at the
empirical evidence I sometimes worry
that there might be some confusion and I
also I'm a little bit uh squeamish about
talking about meaning here because I
think that meaning is a very big
word um so so I I really like this term
incentive salience because I think it's
it's connects precisely what you're

interested in which is salience so the fact that some stimulus is Salient to an organism with the the the idea of an incentive so the the idea that the reason why it's Salient is because it is associated by the organism with some expected reward and that's what's driving the organism's

Behavior behavior and also I mean inert and overt Behavior so paying attention to something might be a kind of of you know uh covert uh Behavior right yeah yeah yeah considered conceptualized as a kind of covert Behavior right yeah I I certainly view it as such a sort of mental action uh yeah exactly down exactly the same sort of um prob you know the same base graph if you wanted to you know the same c and e Matrix if you wanted to uh yeah so there's plenty of things out I mean one is you're absolutely right I think it needs to you know we always need to say we're talking about preferences in the basian belief sense of that neurons have preferences the atoms have preferences but they they have basian preferences they have attractor sets they have so on or at least they look as if they can be described as if and so on and we've had this and we'll we can come to this of course um when I talk about meaning the reason why I do so is because I think the the way that preferences in slave action is is encoded in a hierarchically deep fashion such that you know the certain things that we can claim have uh make our lives meaningful are those

which seem to have this trickle down effect such that they one way of thinking about it is in terms of meta stability you know the the sort of the deepest wells in which the other attractor Wells are kind of subsumed so I've been thinking about this in terms of okay you know maybe my ultimate you know what makes my life meaningful is you know for evolutionary reasons is being a social adaptive member of society now what I pay attention to although it's indirect and and implicit and covert is often a manifestation of that so why am you know why am I writing a paper because I want to get a PhD why I don't I want to get a PhD because I want people to think I'm impressive what and eventually in some sort of Aristotelian fashion you ground out for me in these kind of hyper priors which seem to have this kind of deep filtration into our very low-level attentional schema one question that I've been thinking about as a kind of followup is I think it's very reasonable to say that you know when I'm hungry because it's a very deep rooted prior it kind of has a a standardly high Precision waiting that prior such that if when there's deviation well maybe not maybe it's a um when there's deviation from that prior there's an action policy that grants it high Precision waiting I should say because we're not always hungry or aware of our hunger levels so there's a slight Nuance there um but you know we can just

say that it's a you know for lack of a better word it's kind of an import deep prior MH I think it's fair to say that when that's the case I'm obviously I'm far more drawn to food right of course I'm drawn to the fridge I'm drawn to adverts of food now but I've been thinking is that is it the case therefore that preferences and and that which is afforded in terms of basing preferences or high Precision weight at that moment literally change our perception in such a way as it's sort of like a sapier Warf hypothesis for V for perception or is it the case that our perception is uniform and what ends up happening of course our attention isn't uniform but our perception is uniform and what ends up happening is we kind of overlay uh our perception with the kind of conceptual schema and well also kind of the attentional schema but actually in the sort of fundamental aspects the intrinsic aspects of our experience uh the colors the sounds and so on things remain the same

have you got any kind of intuition on question uh this is a good question because I think uh so so I'm kind of torn on this on this question so so if you're looking at these hierarchical if you're thinking in terms of these hierarchical basan models and you know active inference where there's cascading predictions constraining the activity on the levels below it seems like there should be some form of influence of you know your Global States for example over

your local States right so so being hungry is a kind of a state of deprivation which seems to be a kind of a state which is I would I would maybe label it as a global state in the sense that it pertains to your organismic functions and if it's not satisfied you will eventually die right so so it's kind of a a Glo a state of global importance to the to the cognitive system so to speak because it it's it's the existence of the cognitive system is contingent on on uh on keeping that state within certain bounds so and then there's a question of like whether there's a way in which through as you said saliency maps and attentional modulation and all these things and as you said conceptual overlay for example there's a possibility in which this higher order well let's call it Global State again would constrain the space of possible hypotheses or possible models in such a way that it would have determinate effect on your conscious experience um and I think it definitely can have an effect on your conscious experience overall because if you think of your effective experiences yeah you know we have the term hangry which clearly indicates that you can your effective state is changing and and you can go into a certain mood because you're in this Global state of deprivation and then your your fuse gets shorter and you

get irritable and so on whether it
changes your perceptual

States I think that under some models
people should be committed to claiming
that it

does I am not so sure about that because
this is and and then we're here getting
to the to the second kind of uh part of
this dilemma for me because there's this
whole literature about cognitive
penetrability and whether our beliefs
influence how we perceive the world
qualitatively whether they beliefs
influence perception and from my
understanding um for example the work of
Chess firest Stone and Brian Shaw the
evidence for that is very very limited
yeah so it seems that our beliefs can
for example influence our actions very
easily yeah but they do not change our
perception as readily as uh it might
seem intuitive and so here I would
probably try to be a little bit cautious
and now with one curve yet and that's
that's something that I'm working on now
which is this um the effective ladness
perception

so as you know and as

many people at this point might suspect
being a student of D denet I'm no fan of
qualia and phenomenal properties seem to
me you know metaphysically suspect and I
would like to exercise them or at least
deflate them as much as
possible in our ontology
so I think that a lot of problems
related to ineffability and a kind of
intuitions

of phenomenal States being some how mysterious comes from the fact that in our perceptual experience for example let's take Vision as a as a simple experience a lot of information gets compounded and compressed in a way where we are unable to track all the information that's being you know uh packed into into a perceptual representation by our perceptual systems and so um so you know you said that so one example is that colors appear to be intrinsic properties of objects right to us even though we know that they are actually bundles of complex relational properties that depend not only on the surface properties of the object but also the the source of light uh viewing conditions context our past experiences uh the state of our ocular system and also how our visual system is processing it right so there's a complex Nexus of conditions that obtain outside and inside of us that make colors appear in a certain way H and yet all this stuff is kind of flattened into this perception of colors as properties of objects that are out there in the world and I think that so that's part of the story but even if we if even if I present to you you know the best state-of-the-art knowledge about color perception you will say like yeah but there's something ineffable missing there's this qualitative Gena of of of phenomenal

experience and I think that this is perhaps because in our perception there are Pathways for example to subcortical areas which might be imbuing our perceptual experience with as you said some kind of slight emotional sence right or something like that so what people have called in the literature as micro valances right yeah yeah um so I find this picture attractive especially I find attractive the idea that somehow affect might play a role here but I think that we have very at this point in time at least we have very limited uh very limited evidence for that so this is something that we're doing now in Brussels trying to find out more about uh the effective ladness of of uh perception and also it's worth taking into consideration just finally kind of as a reconciliatory note the fact that you know even if perceptual States as such sensory states are not penetrated or somehow imbued with effective or other types of information from other area brain areas or brain processes that doesn't mean that our perceptual experiences as a complex cannot bring all these things together right so the perceptual experiences are extremely complex so it's not just about sensory properties Drey has for example written a lot about you know how Concepts structure our perceptual experiences so there are sensory properties in our perceptual experience

but it seems that there's much more happening yeah than just the sensory properties alone and that's something I'm happy to concede and I think is highly plausible but the claim that you know I see the the candy bar rapper as brighter because I'm hungry right that's I would I would I would be very cautious about this type of CLA you see that that that's the kind of paradigmatic example right because um I think you're absolutely right we need to bracket off the domains in which perception might be from which perception might be influenced so affect is clearly permeating perception uh and and clearly uh the preferences or the PRI or the hypotheses are affecting that right as you say hunger is a perfect example such that there is this prediction error or free energy spiking with respect to this one homeostatic Prius and that permeates my experience that becomes a uh you know a kind of just all pervasive um yeah uh experience the other thing that clearly in some the wrong is the wrong word pervasive but is over you know intrinsic to my experience is the attentional modulations the the rivets and the peaks of my prioritization and clearly as I say and I outline this paper preferences are shaping that um such that I'm hungry the candy bar is going to draw my attention I think this is fair enough to say more than it would when I'm full right like I had this experience yesterday I went out for lunch I was really I was with

friends I was really hungry and there
you know news you know news agent shops
or whatever selling food and you're in
that state of hunger
drawn to that and then I leave and I'm
like I don't care I don't want to see it
but yeah exactly the point with me does
that actually alter the fundamental
character you know uh the the the
qualities of that experience now I think
an interesting contribution to this
question is
psychedelics because yeah there's
something about psychedelics which if we
take a kind of Rebus or alus approach
loosens those
priors at at some level right I'm not
saying the the sort of preference priors
but the priors over let's say um I don't
know like it's kind of like in dreams
right like you might have a prior that
um I can't sort of dissolve through
walls I don't know like this clearly
there's probably no but something like
that and clearly something in like
dreams and psychedelics that gets
loosened such that in perception I seem
in the dream to be able to sort of blend
through walls and and so on now there
there there does seem to be a relaxing
of a mechanistic prior that permits um
changes in my visual experience in this
case I guess the question here is
slightly different is there could there
be changes also in my sort of preference
priors such that if I was uh if I was
super hungry and then took siloc cybin
would would it literally change my

visual of uh perception or is it just
that the mechanistic part of that is
changed so there's kind of two different
priors or preferences that we need to be
talking about as well can you can you
sorry can you repeat the last part so if
you took silos cybin it would change
what because I think that you lost me
there at the at the very end so I guess
what I'm saying is you know the way
people talk about well the Rebus
hypothesis relax on psychedelics is
somewhat vague about which it's a kind
of narrative account of Prior it just
relaxes and I think KL has said this
himself I'm sure Robin would as well but
there are different types of priors so
one would be a kind of mechanistic Prior
okay what do I expect to appear in my
perceptual experience so I do not expect
uh you know two things to be in one
place at the same time I don't expect
for things to move through solid objects
yeah but then there's also the kind of
what you know you could call phenotypic
or preference priors which is defining
of the organism I expect uh to be
satiated I expect my blood glucose level
to be a certain level I expect my
temperature to be a certain level now
seems to me that what siloc cybin is
doing is relaxing those mechanistic
priors such that you can start seeing
things go through walls and the same in
dreams and whatnot but I don't know how
strong the evidence is that it changes
the preference priors right does it it
would so my thought experiment was if I

was really

hungry and then took siloc cybin I can
imagine it would hijack my attentional
apparatus such that I would start homing
in on all the food

but would it literally you know would my
steak start like glistening would my
candy bar start I don't know I guess at
that point it's kind of hard to say what
what part of the system is being altered
but that was kind of the the fault
experiment okay okay I see I see now um
yeah so well it's clear that

psychedelics can induce sythetic
experiences right so so they can
alter how we perceptually experience the
world

dramatically and uh they can make us
experience you know certain uh this kind
of modality

mixing States right so so you you see
the music for example stuff like that so
so that seems to be a very deep uh
change in in how you experience the
world in a sensory way mhm so that's one
thing um I totally agree with you about
the mechanistic priors and the fact that
this seems to happen uh on drugs a
lot um and that there is yeah that that
that that as just as you mentioned as in
dreaming like they seem to be relaxed so
you're entertaining hypothesis that
would otherwise be

outside of of the hypothesis space that
you could entertain and what's also
interesting about dreams and
hallucinations is that at least from a
philosophical perspective that

philosophers often kind of create this
stman idea of hallucinations as
something that's you know you're you're
suddenly in a state where you don't know
that you're hallucinating and you're
confusing hallucination for reality this
is actually extremely rare given given
current evidence and it seems that
people under the influence of
psychedelics very often are aware that
hallucinating just as sometimes people
in undergoing lucid dreaming they can
enter a lucid dream by re realizing kind
of having this moment of aha of oh my
God this is completely unrealistic yeah
um which is in itself also interesting
because that means that you you you're
relaxing a prior on certain levels of
the hierarchy or in certain parts of the
system but at the same time there might
be aspects of the system that are still
kind of monitoring what's going on and
are able to say this is this is not
reality this is we're we're on drugs now
right um that's one thing and then
finally going to these preference or
phenotypical priors I think that's an
open question one thing that's for sure
is that and I don't know anything about
literature uh that speaks about this
change in these type of Prior in
preference prior
during these psychedelic experiences but
there is a lot of evidence that
psychedelic experiences do change
people's preferences and behaviors long
term right so so there's this famous
effect that people who have

undergone uh a psychedelic experience
are for months afterwards scoring higher
on openness and lower on neuroticism and
and and these kinds of personality
traits which would I would at least
correct if I'm wrong I would ascribe to
a change precisely in these phenotypic
priors where sure you're kind of
changing your behavior or your attitude
towards the world and people yeah uh
because of that experience so perhaps
within the experience it's also possible
now you know whether whether there would
be this permeation of perceptual or
sensory experiences by your preferences
that's a very interesting question and
uh I mean we're in the kind of gold an
age of psychedelic research these days
so I'm sure that sooner or
later someone is going to control the
conditions of their experiment in a way
where they will be able to show that
this happens or it doesn't happen
perhaps right well I guess yeah it's
it's it's super important to unpick
exactly what the hypothesis would be in
that in in in that experiment because as
you say clearly as you yeah I mean
there's overwhelming evidence that of
course you get these longer term effects
and then modulating perception and
conscious experience in interesting ways
but again the question is still there
which is okay what part of your
conscious experience is it uh modulating
so okay you're more open you're less
neurotic it's going to be changing your
attentional schema it's going to be

changing your effective schema right the things that used to piss you off are probably going to piss you off a little bit less you might be less reactive so you might have more space between the the percept and the response but is again I mean I mean the really really the kind of fundamental question here is how much of conscious experience is there beyond all of that you know like how much is color these kind of supposedly intrinsic things that are not um overlays even though we know mechanistically they might be overlays for sure like to what degree do are they are they just part of all of those things I mean the evidence kind of the my intuition is that they're not because I could pay attention to something but that doesn't change it it changes the way that I am living with it right I same with affect it doesn't change I mean I don't know it's hard to say you know no no no no I this is a this is not so this is not uh an argument that has not like this argument has been made before right so this idea that when stay off intuition when you're paying attention to something it changes your quality the quality of your relationship to that thing yeah it's like with a camera you know you're looking through a lens of a camera and you're adjusting the focus but it doesn't change the object or the contents of of what you're seeing right so so so people have argued against kind of predictive processing

stories that attention might be kind of
uh narrowing the the distribution right
making it more precise but it's not
changing fundamentally what you're
perceiving right it's if it was red it's
still red yeah yeah yeah and

[Music]

um uh yeah and and so because I guess
the thing here is it's not changing what
you're saying I mean again like we kind
of have to distinguish between the numo
and the phenomenal right it's not I mean
I don't think anyone would argue that
it's like unless you think that you know
you have some quantum mechanics and you
think there's Observer like Park calling
of that and you Tony the mechanics of
course if I look at my computer I'm not
it's not sudden apart from the very sort
of potential Photon changes I'm not
morphing it into something that it
wasn't sure but phenomenally the as you
say the contents of my
experience I don't know I'm a little I'm
a little L I'm a little less secure on
that front um I mean I'm just noticing
things that I didn't notice
before which in a sense is changing the
content but whether it's changing the
content as it pertains to the object
itself is a slightly different question
right so for example uh the yeah so so
one way to think about it is so so okay
so we have to kind of I think separate
several things yeah so one thing is that
uh my claim and the claim you know that
other that that Joe has made together
with me in the in our paper is

um attention is kind of modulated is is
is corresponding to what what block
labels Act Consciousness right so so so
attention is not supposed to give you a
full account of phenomenal qualities
that's that's one
thing um and I think most people most
higher order theorist theorists agree
with this that you know you can probably
do more than you expect with just this
kind of story about introspection or
some kind of higher order lower order
relationship so people tend to think
that you cannot get any quality postive
information there that's not entirely
true uh and
um uh Andy Clark has actually a very
good paper I think from 2000 or 2003
called access
implies uh access implies quality I
don't remember the title now I'll find
it uh it's a very nice paper is it a
case where access implies qualia yes
that's 2000 yeah uh yeah so that's a
nice paper also that
shows that the distinction between axis
and phenomenal Consciousness might not
be as clear-cut as block and his
supporters would like but just to be
clear on our picture attention is just
you know this kind of probing mechanism
in the netian terms it's it's this
access to information mechanism yeah and
now so that's one thing uh now the
second thing is as you said that um that
the the claim about whether it changes
the content drastically whether paying
attention to something changes the

content of your experience and here again we have to be very careful because something can change uh your experience can change in some qualitative ways without the content changing per se so so Drey has this very nice example where he Compares two situations one in which I show you a piece of paper on which I drew 30 dots and I you know I just kind of show it to you like this it off and then in another case I show you the same piece of paper but there are 31 dots so technically speaking technically speaking the number of dots has changed between the two pictures so the content of your experience has changed Technic at least that's the point that he would like to make in my in my view unless you notice that there's an extra dot the content of your experience has not changed right and it's only through sharpening of attention Could you actually notice that there was a difference right in your in the between the between the two percepts so so so I think that there are these cases where indeed uh paying attention can change the quality of your experience even though it might not change explicitly the contents of your experience interesting that's an division right because I think most would view those as isomorphic they would view contents of experience and for and the phenomenal qualities of experience as isomorphic yes so so I think it's useful to maybe

think of oops I think I'm getting now
sun is shine into the
office uh so so you might think about
this in terms of modes of presentation
of content right so attention seems to
attention is not only picking out what
aware of but it can change the mode of
presentation of what you're aware of to
some extent just as I said earlier with
the metaphor of sharpening the the the
focus on the camera lens yeah the the
the subject in the picture doesn't
change but what's changes is kind of the
way you're capturing the picture with
your camera and I think this is this
this might this kind of talk of modes of
presentation might be useful as a as a
useful kind of intermediate step between
the the content itself so what is
represented yeah the way it's
represented right um this mode of
presentation those that that distinction
might be might be useful here as well do
you think there's a strict do you think
one can draw a strict line there between
contents and mode is that do you think
that's feasible I mean is there a kind
of taxonomy of modes in which things can
be uh perceived or can my mode in some
ways become opaque and become a content
I think the latter can probably happen
so so I there's there's not many people
in philosophy of Mind these days talking
about M of presentation so charers is
what is I guess the the main figure when
it comes to that uh Hilla Jacobson talks
actually about something similar when it
comes to effective uh effective coloring

um and uh some other people do as well
but it's a very small group of people
these days that talk about modes of
presentation as distinct from contents
but I definitely think that there is a
there is a there there are cases where
we conflate the the way we experience
things or the way they are presented to
us in experience with what they are so I
think I think so hila Jacobson has for
example a very
nice uh example that she uses in her
paper as part of her argument which is
that the same object the same food item
can look delicious
and disgusting to two people yeah and so
what seems to them they are looking at
the same thing right so so the sensory
properties do not change presumably or
we can even assume that the sensory
properties don't change between their
two
experiences but the effective mode of
presentation of these two sensory
experiences can be dramatically
different right so so and and a simple
example that I used in my talks
of hijacking Hill's argument is is a
piece of licorice because people often
think that you know people can have
strong reactions to different food items
and licorice is a is a piece of sweet
that almost or Marmite would be another
example those are two things that people
have very very strong opinions about and
they have very strong reactions to but
the a jar of Marmite or a stick of
licorice are not by themselves repulsive

to us people just perceive them as disgusting or delicious and you know they can have this kind of different ways of engaging with that content so maybe that's another way of thinking about mode of presentation as as this kind of way in which you engage with the with the content of your experience right what's uh Shakespeare there's nothing good or bad but only thinking makes it

so yeah yeah that's that's Hamlet um yes

I mean Shakespeare I guess wasn't thinking about whether yeah I guess uh I

would have of course I have no issue

with the idea that licorice might

look words Fair here might look

disgusting or look attractive to people

but I guess the question is do they look

disgusting and look attractive to people

are we overlaying that or is it

suffusing our I mean even if it's not

overlaying it might be suffusing our

experience the question there as we were

talking about is are the sensory

properties being altered but I guess I

mean I guess that's a I guess it's an

experimental question ultim I mean how

you could how you could do that without

qualitative report I mean that that

which might be quite limited I'm not

sure because again I mean language fails

us here so this is what we're interested

in here in Brussels right now and and

one way you could think about it is well

do you see

changes uh in the sensory processing

areas right when you're when you're

putting people in a brain scan right do
you see changes in activity depending on
the veilance of the stimulus that
they're showing them so so a common
stimulus used in experiments on uh
affect are smiling and unhappy or
frowning faces or or angry faces right
so you show them two types of faces they
distinguish whether they are happy or
not and then the question is where do
you see the difference between
the happy and the and the frowning faces
right is the is the difference
represented the difference represented
in

some area of the brain that's associated
for example with processing value yeah
uh some kind of one of the frontal areas
or is it maybe or whether we can find
the differences already on the level of
perceptual processing so this is
something that we're aiming
to uh to test here and there is there's
kind of the the
available uh evidence is kind of
equivocal I would say so there are some
papers that show differences in
perceptual areas there is also some work
that's been done in Nicholas krias
scores lab which seems to show that
there are
actually it's not that the differences
are represented in sensory modalities
but there is a sensory modality specific
coding for the veilance of the stimulus
so you could think that there's a
universal code for Value in the
brain but in order to compare where you

have to bring two different stimuli
right so if I ask you if I show you
a happy face and a monetary reward
something like that if I give you two
two different types of stimuli if you're
comparing if you have to do some kind of
a trading off between them um you have
to bring them into some kind of a common
neural

currency yeah so there's K ctis lab uh
has shown that there might be this
Universal value coding but before items
are brought into this Universal value
coding for comparison there's also a a
modality specific coding of value uh in
the brain that is not sensory per se but
there is there are differences in how
values are

coded uh depending on the modality from
which the information comes which would
seem to be this kind of intermediate
state where your perceptual experience
might

be uh effectively Laden even if the
sensory properties stay the same because
what's there's something that's being
added you know on top of sensory
properties it's not changing how the Hue
and the brightness and the color that
you're perceiving but that there's
something else that's overlaid on that
as you suggested earlier that is
changing the character of the overall
perceptual experience and this is
precisely Hilla Jacobson's View and also
Frederic de Vino has a recent paper
where she also adopts a view like that
so much to read so much to read I'm

writing down notes I mean yeah I mean
I'm very impressed I mean it's funny
that we still haven't resolved these
issues uh since 1983 when Jerry Fodor
wrote modularity of the mind I mean he he
he would if he was still here he would
have loved to um he would have loved to
have uh listened to this conversation
because I'm sure he'd have lots of
opinions but perhaps or he would be
maybe he would love to or maybe he would
be extremely frustrated you
know yeah well I'm sure I'm sure again I
did I never met Jerry of course but I'm
sure he would uh whatever it was I heard
he was very he was an opinionated man I
mean for good for good or for bad okay
cool um that was that was that was super
interesting I mean I love that I read
modularity than M when I was at Uni and
loved it I mean I loved it as an
intellectual Endeavor I mean I had no
skin in the game but um I thought it was
I thought
I also think there's a lot you can learn
about writing clearly and
precisely from from Fodor and from Drey
so I've never read any Drey I mean he's
come up over and over again I had a
little foray into I read one of the like
radical and activist books
um okay can't remember who wrote it and
there's a lot of sort of dryan and
milikan sort of critiques as in they're
critiquing them because they don't think
that any you know only represent the
only representations are those which are
socially scaffolded yeah oh so it's the

second book by uh

Dan May yeah and Eric mine yeah the the basic Minds evolved I think yeah I read I read yeah I read one of them um and again like it's not really my argument um but it was interesting enough I should read milikin and and dry but life is life is short high but they're highly recommended so so I really um knowledge and the flow of information especially if you're working on predictive processing and these topics I think it's a really beautiful book that probably is the first philosophy of Mind book that took information Theory seriously uh even though there are some distortions and he at some point develops his own kind of you know version of of of Shannon's communication Theory kind of Mong view in the end but but it's a beautiful book and it's an extremely funny fact that that book was published in 81 I think so 30 something years after you know Shannon's commun mathematical theory of communication became a thing and this kind of shows you how long it sometimes takes for philosophers to you know catch wind of what's happening in science did um this is a kind of historical question did Chomsky ever reference Shannon was try do you know if trky was aware of Shannon I'm I think it would be strange if he wasn't given that he was you know where he was working and that would be surprising to me however I don't actually know so so I've read

Chomsky uh but a long time ago like his linguistic writings a long time ago and I mostly the last things I've read were mostly about um about mental representations oh really yeah so so he has a very very deflationist view of mental representations you can probably find him you can also Francis Egan in some of her papers okay takes his minimal view of what she calls erat's representationalism as a kind of uh Target yeah I did uh I did Linguistics as my undergrad oh okay so I I read Chomsky from you know um syntactic structures all the way to minimalism uh but this was all very sort of biological Linguistics about in you know the question of whether Linguistics there you know a laid an innate linguistic device or things like this so slightly different but uh yeah I was just wondering because obviously he was writing in you know the 60s 70s so yeah yeah okay uh on the question of representations I mean actually it's funny because both people I've had on who are expert in representations you and Alex Kefir have both recommended me dry and I'm Alex's work on representation is great so I am a fan of his work you I really liked your uh moderate predictive processing paper it was challenging because um I feel I it kind of made me realize like oh I've been taking the vanilla predictive processing because I'm not personally in my own writing I'm not wedded to any particular

architecture whether it's discrete State space continuous State space I take the vanilla position I run with it and so on but it made me think like oh there's so much more for me to learn exactly about predictive coding predictive processing schema well I found really interesting was this idea of well this ambiguity more broadly in the literature about what is being represented at different layers of the predicted processing hierarchy so you say and can quote um there's an ambiguity between levels which perform the function of standing in for features of the environment uh and those which act only as meta representations by standing in for the features of levels

Downstream so there's this idea that obviously the high you know the higher you the higher layers are predicting I mean all layers are predicting but you know they're predicting the layer beneath them and the prediction area is rising up and there's kind of this meeting in the middle and this Hier this horizontal explaining way of prediction error but I thought you this is a really nice observation which is where does the external world come in right because there seems to be this equivocation where we say okay we're predicting the external States across the mark of blanket but then in this predicted processing scheme we're actually only predicting the ne the layer below and so there might only be one layer which is actually receiving the sensory input so

where where have your where you know
this was 2017 you wrote this paper where
have your thoughts now evolve to with
respect to the architecture of
processing uh yeah so just on that quote
right so so just to be clear about what
I was kind of attacking there so so what
I was attacking is this ambiguity
between in in an in how people answer
the question about what is the function
of of a layer in a in a predictive
coding hierarchy so is it representing
the world as some people will say or is
it predicting what's happening just
below on the level below and a lot of
people when I first encountered
predictive coding and uh it was very
informed by Mark SPAC who was my master
thesis supervisor in edinb and he the
way he presented it to us initially
was uh in saying that the brain is just
minimizing the the story is that the
brain is minimizing prediction errors
and that's all it cares about and then
it turns out that the best way to
minimize prediction errors is actually
to represent the world right so so so to
kind
of have the the the brain become a kind
of distorted of course but a mirror of
nature in some ways and that this is
this the reason why predictive
processing offers a a kind of a story
about the organism building up its own
model model of the world is because by
building the the the process of building
up your own model of the world is just
the best strategy for long-term

prediction error

minimization um and so my problem in

that paper was that different people

want to clink to different

interpretations of what a single layer

in the hierarchy is doing depending on

what on what serves their

wider uh wider philosophical Pro object

so people who are

anti-representationalist will say like

oh you know the function of a layer is

just to predict what's happening on the

layer below uh and other people will say

like the function of a layer is to

represent the regularities in the

environment at a certain level of

abstraction and at a c in a certain time

uh

window and both of these interpretations

can be true according to predictive

processing it's just that which one

you're try you're emphasizing seems to

depend on your uh external let's call

them or at least external to the

framework philosophical assumptions and

convictions and this is why when I wrote

that paper it was already becoming

apparent that for example

Carl uh is uable to different

interpretations of predictive processing

or predictive coding or active inference

whatever you want to call it depending

on who he's coauthoring a paper with so

so if he's coauthoring with an an

activist he will take a more an activist

stance where he will uh try to avoid

talking about internal representations

or or the or building up an internal

model of the environment yeah when he's writing with someone who's uh coming from a more cognitivist fodorian uh persu with with a kind of fodorian persuasion then he will lean more into the representational talk so it seems that uh that there was this ambiguity uh about what's what's happening in the framework and today I think I'm kind um so I used to think that maybe there's a definite answer to this question which interpretation is correct today I'm actually thinking that uh the semantics maybe are kind of disconnected from the from the formal layer MH so I have this I I'm I'm at least when it comes to predictive processing I think there is no definitive answer whether we should think of the generative models for example as representations or not I think that this will depend on your other commitments about what a representation is for example uh on the wider goal of what explanatory project you're engaging in I think those are the things that will actually determine which interpretation you will apply to predictive processing but I'm not uh really committed to to either of the interpretations and um this goes a little bit back to attention in the sense that I think what you attending to in the predictive processing story might influence how you

will want to

interpret uh the framework I'm sorry if

this is kind of a oh no I love it not

satisfying answer you know it's maybe

I'm the same I'm the same uh despite

being a host and and I feel like part of

being a host of active inference podcast

is in some ways you should be wedded to

certain ideas I'm pretty ambivalent

about or not even ambivalent I'm open

right I'm epistemically open I don't

know um that might frustrate some people

but I don't know uh and also this isn't

really I don't really have an area of

expertise yet so so I appreciate the I

appreciate the epistemic humility I mean

it's an open qu I mean I mean

fundamentally one of the issues is no

one really knows what representations

are I mean like what what what the I

mean as you say all philosophical the

all theories have their miracles they

have their axom right so are you a dreki

and are you a milikan follower are you

you know do you think that there needs

to be a you know a strict isomorphism

between that which is representing and

that which is represented and so on and

so forth I wondering whether even if we

park the sort of what is being predicted

whether it's the lir below or whether

it's the world

what are the sort of unifying features

therefore that we've come to in terms of

a predicted processing landscape so for

example there's this idea that the

priors of one level are the posterior

furnished from the lay above yeah now

what does that really mean is that is
that what the is that what people talk
about when they're talking about
predictions coming down the
hierarchy and and I guess the question
here is and I said this to Alex as well
I think there's a real I mean I struggle
visually conceptually to really imagine
a predictive processing hierarchy
because I always view it as in time
right dynamically using such that you
have these I mean at a very vanilla
level fine prediction eras coming up
predictions coming down and they get
canceled at some point you get a
hypothesis woo the world appears like
great but when you actually start
thinking about okay mathematically
formally what are the architectures that
we're setting up in place it becomes
very tricky for me to imagine because I
think of a layer that is simultaneously
predicting and being predicted and
explaining away prediction error so just
at a very broad level is there for
myself and for other people listening
and watching a way of like formally
disambiguating what might be happening
or should we just say this is some
dynamic system and you can kind of get
by with the vanilla approach I think
you're putting me in a position where
which is you know way above my pay grade
here
no but
um uh yeah that paper you that paper you
do cover quite a lot of technical
round a little bit but I must say that

also I'm not entirely sure how well you know my technical grasp has so so whether I was that correct there when about in the things that I was writing did anyone send you an angry email no no never never but this this I I'll answer that question your question first and then I'll tell you one anecdote about this paper so um so yeah so you're right that there's a lot of variability when it comes to how you can implement the general predictive coding scheme and uh Michael Spratling has a very nice paper from 2007 called an overview of predictive coding algorithms where he compares different first of all he gives a kind of a brief historical overview of linear predictive coding and where it came from and then he talks about the three uh yeah the three major varieties of predictive coding as applied to hierarchical networks which is ran Bard's original model uh Carl's uh free energy minimization model and his own predictive coding through biased division modulation I believe that's his the name of his algorithm so he compares these three algorithms and Michael Spratling is a is actually a guy from London who who has done uh a lot of work on modeling uh neural populations using predictive coding and building very large neural networks performing different cognitive tasks by through by implementing implementing predictive coding and so uh

one of the problems that that paper highlights is that because there are many different ways of doing predictive coding you know it's not stable from one algorithm to another what the prediction error is right and you know what is propagated upwards what what is propagated

downwards even how to delineate a layer of the hierarchy well quite yeah uh so so there's a lot of and that's what I also touch uh that's what I also touch on in

that 2017 paper moderate predictive processing so there seems to be a number of possible ways of doing predictive coding and depending which model you will or algorithm rather I should say you'll pick there will be certain differences and what counts as what so in the end when I was writing my PhD I actually decided to go to a higher level of abstraction

and I looked at the work of Iris vanroy and Johan kout from radbot University in Nan in the Netherlands so they have a bunch of papers in which they formalize coding on a more course grained in a more course grained way as uh by using basically basian graphs and imagining each layer of the hierarchy as a basian graph with certain dependencies and uh you know connections between variables so in their model they assume that each layer of the hierarchy very much like in fistance models each layer of the hierarchy which will have a set of hidden

variables and a set of prediction variables right and these uh these these variables will correspond to uh yeah so so these two types of variables will correspond to kind of the the latent variables and the so-called observed variables in a way where the the latent variables are the are the hidden variables that get inputs from The observed variables on the level above when predictions are gener predictions are

generated uh and this will be integrated and then at the same time will'll issue predictions so so will influence the state of their their own observ observed variables and these will kind of cascade downstairs so so they imagine this kind of simple probabilistic graphs that are connected uh one to another and they kind of operate in this kind of two stages right so they either do inference or they or they um observe what's happening and I think that's a very uh General this is probably the most General way in which you can think of predictive coding or additive processing because it leaves out error units so they don't talk about how errors are propagated into the system where the where the errors should be connected to to things and stuff like that and that's I think really an important abstraction because the way errors are processed are dramatically different between different predictive coding algorithms and their very abstract and

very general way of for normalizing predictive coding has the one crucial benefit that it abstracts away from different algorithms so that you could still capture the general similarities between these algorithms without talking you know or getting bogged down from my perspective into the minutia of how each of the algorithms is implemented exactly now there's a trade off in going via that route because by going for a higher level of abstraction you're of course losing detail and if you're losing detail sometimes you might actually find out that you cannot disambiguate certain claims about predictive coding on that higher level of abstraction so people who have for example very demanding views of mental representations as structural representations that are capable of action guidance and error correction and stand in some kind of homomorphic relationship to their targets in the environment well it becomes very difficult to disambiguate whether these abstractly captured predictive coding layers are working as structural representations because you've left out so much detail on the table that you're just not able to exactly adjudicate whether they fulfill all the criteria or not right so so the fact that an abstract abstracted and idealized description doesn't fulfill these criteria that these people will have for mental representation doesn't mean that a particular model like spread

links Network or distance network

wouldn't fulfill these

criteria uh on its own when it's

properly implemented with with a greater

uh level of

detail that's again not a very

satisfying answer I feel no look I no I

mean I'm glad personally that it other

people find it confusing

and that I can go to spratling or Carl

what remind me of the two Netherlands

the two Dutch people who you were

influenced by Iris Iris vanroy Iris Iris

the way you would write it in

English van and Roy is rou i j yeah and

Johan

kout okay perfect thank you so highly

recommended uh very good

very good Visions no that this is I like

because I I think part of um part of my

role here is to point people in certain

directions right um so I'm seeing here

cognitive cognition and

intractability so they they they they

worked a lot on cognition and

intractability I think um the paper I'm

thinking of or the where the

formalization appears is something like

to be precise the details don't matter

oh there's free energy minimization and

information G the devil is in the

details might be that yeah that's I

think a commentary on Carl's paper where

he mixes discreet and continuous

variables so I'm thinking of uh 2017

Johan kout Harold beckerling and Iris

vanroy to be precise the details don't

matter unpredictable processing precision

and level of detail of predictions that's in brain and cognition okay fantastic well thank you for that uh it's a paper that definitely influenced a lot of my thinking about uh about attention in the in the excellent I mean it's a sh framewor it's a shame I wrote this paper before um it's difficult to keep up with all the literature because also the literature has ballooned so I started my PhD in 2014 and at some point I was just getting like there were 20 papers coming every month on the topics that I was working on just from people working on predictive processing so you know there were in five months I had a 100 papers to read which is kind of yeah yeah unmanageable right so so so um some point that literature just exploded I'd like to know your anecdote about this so so the anecdote since this is you know a public facing podcast or at least I hope hopefully the the academically facing podcast the the story about this paper is that I wrote it uh I guess it came out in 2017 It Was Written somewhere between Yeah March 2017 it came out so it was written mostly in 2016 so the early days of my uh PhD in fact and um and it was a little bit you know it was hot on the heels of Andy Clark's radical predictive processing which is which the title is supposed to be a reference to and the debate between radical and conservative predictive processing so precisely

uh representation light or
representation heavy interpretations of
predictive processing and I made a
terrible mistake when naming this paper
because it's called moderate predictive
processing and to me at the time it was
very clear what the title refers to and
what it is about yeah um and I thought
you know it appeared in this collection
of papers philosophical papers on
predictive processing edited by Thomas
metzinger and Vana visa and so I thought
like everyone will know like this will
be obvious that if you're reading like
if you're Googling stuff about mental
representation and predictive processing
people will just find this paper no
problem and it actually turned out that
because the title wasn't immediately
informative people just didn't know what
this paper is about
so I was meeting people in you know four
years later or three years later who had
no idea that like this paper came out
and they were I was sitting in talks and
people were talking to me
about the these things and they had no
idea that this paper came out and at the
time it was one of the very few papers
about mental representation and
predictive processing but because it was
just badly named just people it didn't
show up in searches so I would advise
and I would advise I would advise
especially Junior people to to be
mindful of how you title your work yes
because that actually can change the
discoverability of your work quite

significantly yeah they prob they
thought you were making a political
point they thought you
were H very good yeah no you're
absolutely right just like fill the
title with buzzword right just free
energy free make sure the word free
energy in there or active inference um
yeah just yeah well discussing public
facing work people are going to be
surprised if I don't mention the
emperor's new mark of blankets but I
don't want to spend too much time we've
been going for a while as well and I'm
aware youve probably got dinner I've got
dinner and so on but it's sort of your I
mean again this is not this was not your
solo project um this was actually I
don't know whether did jelly brunenberg
lead this or was his name just first up
uh so the way that project came about is
because y came to bom where I was at the
time just fresh after my PhD and he gave
a talk about Mark of
blankets uh which is also by the way a
topic which placed prominently in the
previous paper we discussed in the 2017
paper about mental representations where
I talk about the insulation of the mark
of blank with a mark of blanket between
different layers of the hierarchy so
after my after my after I finished my
PhD y came to B and he gave a talk with
his where he was just trying to
understand the difference between Mark
of blankets as they're described in the
literature on probabilistic modeling and
and in literature on basian inference

and how they are used and described in free energy modeling and so he was very confused uh I was very confused and so that was a paper that he was that was a talk which where he credited it to Manuel baltieri and himself that was something that that was a project that they were discussing and working on he gave the talk and I later told him like y I love this project because I've been trying to figure out how they understand Mark of blankets for a while now and it was something that I kind of bumped into during my PhD and I it wasn't entirely clear what is the correspondence between the way that friston describes Mark of blanket and and how Pearl describes Mar of Blancas during my PhD I bought Pearl's uh 1990 book I believe u on probabilistic inference in thinking machines I don't remember the whole title but it's the book in which he which is often credited as the source in which he introduced the notion of a mark of blanket so I bought the book I read the whole book and I still didn't understand how it's supposed to gel with the free energy stuff and then for half a year or something like that you know I was like okay Y and Manuel wrote the paper it's all good I I I don't know half a year more probably passed and and I talked to I was talking to Joe and we were talking about something completely unrelated a paper that Joe was writing at the time which also had some probabilistic modeling elements in it and we talked again about Mark of

blankets because we were talking about
this topic and
uh I said hey I got a great reference
for you because I'm I'm I wanted to
write something about this but Y and
Manuel I think already wrote a paper
about this so I'm going to ask y if he
has a draft and I'll send it to you and
so I contacted Y and it turned out that
they haven't written the paper that the
paper kind of you know they they they
didn't have the time to to figure
everything out it kind of laid in the in
the in the in a that was locked in a
closet so to speak where it was kind of
Sitting Waiting for a better time and uh
and somehow we just proposed that hey
maybe we could collaborate on this maybe
we could help you bring the paper back
so we discussed this stuff and
uh uh we started writing a paper then we
started thinking okay so where should we
where should we try to pitch this paper
like what Journal are we writing for and
I think it might have been me although
I'm not entirely sure who said you know
hey how about BBS like because it's it's
it's it's getting a little bit annoying
these people in the free energy
literature they're stting they're
starting to claim that a bacterium has a
mark of blanket a brain has a mark of
blanket a person has a mark of blanket
the Earth has a mark of blanket right
there are all these papers about the
biosphere or the climate as a mark of
planetary Mark of blanket and so maybe
given how important and how prominent

Carl's work in computational
neurosciences these days maybe we can
pitch it to BBS and so we wrote a whole
draft we we we we sent it to BBS and the
rest is kind of History so we had yeah
so okay eight reviewers I can tell you
about that yeah that that that is crazy
I've had
three well that's a lot eight is I mean
obviously that's my I haven't got
anything out yet so these are all in
process three is three felt like a lot
eight is outrageous um well the good
thing is that the more reviewers you
have the kind of the more of a Arena it
becomes where you can try to pitch them
against each other in your responses
right so you can and and we did have in
in our reviews we did get you know
reviews where One reviewer
said cut this out this is irrelevant and
the other reviewer said oh I find this
bit really nice I'm really happy that
it's there so right well okay so I mean
this this paper I mean this I feel like
when I read this paper I view it in
concert with um The Mark of blanket
trick so I I view those the these as
kind of coupled uh critiques of active
inference or predictive processing
critiques I actually think is H it
probably is a critique um I actually I
read these as more like specul they're
not speculative in a derogatory way but
they're calling into question right
they're saying okay let's let's get
clearer on the terms that using so for
example a distinction that you draw here

is between a so-called Markov blanket and a Friston blanket precisely what what just in sort of layman's terms without getting into you know the granularities of Pearl and I'm going to ask who did the math in that paper because I'm kind of amazed you know I looked at the list because sometimes you see there's like five philosophers and a mathematician and you know they've done all the formula formula and I was like oh no there's four philosophers here I'm like this is really cool that they've also done this um without like getting into the granularities of the variables what is the difference between a Markov blanket and a Markov blanket like in short yeah okay so in very short terms without uh going into the details of you know Pearl's work so so the main idea in graphical models is that sorry in Bayesian graphical models I should say In Bayesian directed graphical models is that you have a number of variables that are related by some dependency probabilistic dependency relations and that's usually expressed by there being kind of arrows between the between the variables right so you have variables that influence other variables and then if a variable is dependent probabilistically dependent on another variable there be there will be an arrow that connects them pointing kind of the in the direction in which the influence go so if there's you know variable A up here that's modulating the behavior of variable B down here there will be an arrow pointing to to from

from one variable to another and that will mean if there's an arrow connecting the two variables it will mean that the dependent variable is conditionally sorry I shouldn't say conditionally but it's dependent on the on the on the variable which is influencing it and so a mark of blanket in very simple terms is that if you have a graph structure where there are different variables connected and dependent to each other in different ways uh a mark of blanket identifies a set of variables which render some variable of interest a variable that in which the state of which we're interested in it renders that variable of interest conditionally independent from other variables in the model so basically it's a set of variables uh that render the variable of interest conditionally independent from the rest of the model so what that means practically is that you have some variable that you're investigating and you're interested in learning everything you can about that variable that means that all you have to know about the rest of the system are the states of the variables in its Mark of blanket and that because that Mark of blanket is kind of shielding that variable from the rest of the system so if you know the states of the mark of blanket variables you know everything there is to to that is needed to make an inference about the variable of inference and that's basically all that Mark of blankets are right so Mark of blankets are

basically uh a way to partition the variables in a directed graphical model probabilistic graphical model um partition them in a way where you will be able to effect efficiently and effectively do inference about some variable in that model um and so what's important about this is that the the the original formulation of a mark of blanket by Pearl that doesn't ascribe any kind of special relationships there or special functions to the to the to the variables in the mark of blanket I mean I maybe yeah like all of all of the V all of the variables in the mark of blanket they can be connected in different ways of course probabilistic to the variable of interest but they don't you know none of them are kind of more important than the others so to speak um and they don't play you you cannot learn anything about the functions they play in the model just from their positioning in the in the mark of blank and now here's where the innovation of Carl friston comes in which is that Carl friston has partitioned the variables in the in the Mark of blanket into the into so first of all he partitions them in in two ways so he calls them the internal variable right uh and the external variables so so he assumes that there's the mark of blanket is kind of separating or shielding the internal variable which will used to be our variable of interest in pearls terms

from the rest of the system which is external to the blanket so to speak and now the mark of the the second Innovation is that apart from talking about internal and external variables he also starts talking about uh ER he starts to talk about sorry and now I have to actually refresh my memory because this been a while since we wrote that paper I assume the sensory and active St yeah exactly I was just wondering what is exactly the term so not sensory act sensory States and active States right so there are the the states in the so variables in the the mark of blanket itself can take a kind of a sensory function or they can take an active function and the difference between these two is that some states some variables in the in the mark of blanket will take information from outside of the blanket and these will be the sensory States and there are some uh variables in the mark of blanket can influence what's happening outside of the blanket so they have kind of these istic arrows in the model coming out from from the from the blanket and influencing external States and these will be active States MH uh and this is this this is it seems at first as a kind of a innocuous you know distinction yeah sure so so what you're telling me you could say so you what you just told me is that like what Carl noticed is that some variables in the blanket are influenced by external variable and some variables in the

blanket are influencing the external variables right and that would be all fine good if uh it wasn't that this distinct this partitioning into active and sensory and internal external wasn't pay playing an important role in in active inference in that it is taken to delineate a minimal active inference agent and this is kind of the leap from from Del from from uh calculating or delineating a blanket and going into it giving it an interpret interpretation of an agent that has some internal States and is interacting with the external environment via its perceptual and active states that that move from one to another is actually a huge leap because in Pearl's original formulation of Mark of blankets you can specify a mark of blanket for any variable in your probabilistic directed a cyclical graph so as long you every variable in the model will have its own Mark of blanket and these will be partially overlapping of course in terms of the variables that make up the blanket but what's important is that no variable is kind of privileged in relationship to the other variables in the model in terms of uh it's it's how it's treated so so every variable has you know a mark of blanket is just a way for Pearl it's a formal tool for learning something about the dependencies in the model nothing more and now and doing inference of course over the states of Interest so it's kind of just a tool that's deployed in a purely pragmatic way and in fistance

interpretation it becomes something more because suddenly there's this whole layer interpretive layer that's added to the mark of blanket formalism which implies that you know variables of Interest are somehow and Mar and blankets are constitutive of agents and it seems to and then implies that you know there are these agents and and the way that friston uh often kind of formulates his models there's an agent that's interacting with the environment and so the the the the interpretation of the mark of blanket under the free energy principle and in free energy models doesn't seem to be uh doesn't seem to be uh sorry seems to privilege a certain you know certain way of looking at what the model is supposed to be doing so it's kind of a built-in assumption that we're modeling an agent and this agent is doing something right in our case minimizing free energy right yeah yeah yeah so yeah this is this is interesting this is a conversation that's come up a lot in this podcast I wonder whether as a result of papers like yours um as well as a mark of blanket trick as well as Mel andrews's paper vanz other I wonder whether there's been a general attitude shift towards a well at least a recognition of this issue so I I've read the the the response as well and the commentary to the responses and you draw a distinction I think this is interesting between literalism

realism and let me just check I don't
instrumentalism realism and literalism
and I like there's a graph I I don't
know maybe it's just because I like
graphs there's a nice graph where you
sort of put everyone on the Spectrum um
indeed where what can you in brief I
mean I'm sure people okay so like
instrumentalism I'm sure in part you
know in short we can say okay the of
blanket is a tool for us scientists uh
but it's not actually making any claim
to ontologically say what's going on in
the world and then realism or sorry
literalism would be well no the mark of
blanket literally is a partition in the
physical world what is the addition of
realism there so that's a kind of the
the Spectrum and then you got realism in
the Middle where how does realism carve
up that
space well so I guess uh
the the way I would think about realism
is that
um that
uh that they they identify so so the
realism would be kind of a position
intermediate position where where you
say well there are cases where in which
blankets do align with or are can be
used to identify real features of the
world right so so are are set to
identify boundaries in the world uh the
difference to literalism is that in this
interpretation a mark of blanket is
still a tool kind of a a a a way to to
use a way to capture a real difference
in the in the world right uh but you

cannot you someone who's a realist at least in my understanding and maybe my coauthors would have a different understanding also time has passed so we opinions but my understanding here would be that literalism is this type of claim that at some point appeared in the literature that you know an organism has a mark of blanket and I think that if you say an organism has a mark of blanket you're committing a kind of category mistake yeah because Mark of blankets are properties of are formal properties of models MH uh and not of organisms and so a more moderate position from that is to say well there's a boundary there's a real boundary in the world and we can delineate that boundary using represent that boundary using a mark of blanket right yeah and that under some conditions and like this is really important because I think we got a lot of push back for that so we we don't we never kind of argued against The View that you can create a model system where under some conditions there will be boundaries in this emergent boundaries in the behavior of the system such that you can model these boundaries with Mark of blankets effic effectively and I think this would be a kind of a realist position what we kind of argued against is that some people have claimed in the literature before we wrote that paper that the all this talk about agents uh

having boundaries separating them from the environment and having internal external States and so on all this talk Falls naturally from the math from the math of probabilistic inference and variational free energy minimization and what we wanted to show is that that's not actually true if it falls from something it falls from some extra assumptions that you bring into the picture it wasn't you know it's it's a it's a it's a result of assumptions that you build into your models sometimes and that's fine if you if you're transparent about that there's no problem with that but it's not that you know you just have a a system where you can delineate a probabilistic graph you you can partition the variables and suddenly you show that oh my God there's an agent in there right that's that's not it just doesn't follow in this straight way and so I think this would be the the clear difference here between instrumentalism also in literalism where instrumentalism is you know just being as non-committal as possible saying we're just doing modeling we're not claiming that what we're modeling needs to correspond in any uh in any truth preserving way to what's out and there in the world then there's realism where we say we are doing modeling but we do claim that we're tracking something real in the environment using this formalism and then literalism is saying the model is what is being modeled right the medium is the message

so to so to speak so so the blanket Mark
of blankets are something that organisms
have that are boundaries in the world
that we can talk about boundaries in the
world as Mark of blankets being out
there and that's just nonsense in my
opinion at least so okay okay so yeah so
this is the point right is it that's the
last point that that last point I think
is the really critical one

uh why is it

nonsense that a coupled agent Arena
relationship has a mark of blanket like
what what is the part so there's one
argument which is that it's a tool so it
started off being a tool and it would be
as you say it's a category mistake to
confuse the map with the territory in
this case so to confuse the tool with
the actual on the thing that exists now
that itself is not an argument against
the thing that exists

the thing exists being isomorphic to the
map the map being the of course it's
just saying it's just saying hold on
right be careful right it's just
precaution now it takes a further step
to go well actually I think that it's
nonsense that the map is the territory
and I think this goes a little bit to
what um V Vicente rajer and his
colleague spoke about that there are
certain systems at least in his
perception that it's not just the case
that the tools that we bring to bear on
these systems

um you have to be precautionary when you
apply them and say that there's

something onto logic happening he's saying
that the tool is insufficient right like
whatever the tool is doing it's in no
way M there's no way we can even it
doesn't even make sense to say that
something like that is going on in the
real world and he has the example of a
candle flame right and this idea with
the candle flame is that the flickering
of it implies that this Mark of blanket
is kind of Shifting in and out of
existence there are further arguments I
mean they won't really pose as arguments
but Andy Clark's things about
metamorphosing Mark of blanket your
own Mark of blanket super interesting
papers Mike Le's work on butterflies and
cater like questions of identity very
philosophical questions but I'm kind of
curious about how you go from saying
okay it's an instrumental tool now I
could be agnostic about how much the map
should be conflated with the territory
but it's a tool and it should be treated
as a tool and we have a different name
for what's actually going on in reality
and it might be the same thing but let's
use a different name so these are just
very scientific questions mhm it's a why
is it why did you take the further step
to say actually no it's not just that
it's a tool and we shouldn't apply it
for scientific reasons but actually the
to it it's literally not the
territory uh yeah so
um okay
so I I think there are two things I
would like to say about that so so first

would be like an intuition pump right so
so me saying that I have a that I have a
mark of
blanket would be something as ridiculous
as me saying that I have number
two what does it mean that I have number
two right so so if if you like if I tell
you I have number
two you're going to say in what way in
what sense do you have number two like
do do you mean that you you possess the
concept of number two or that you can
you know that you can calculate with
number two or you have something that is
uh that is shaped like numbered like the
Arabic numeral or I have two items that
essentially you know that's like what do
you mean by that and I think that so so
the the intuition here would be saying
that you have a mark of blanket is very
similar to to claiming that that you
have number two like it's you it cannot
be set without further qualification so
in this sense it's nonsense because you
know you have to you have to you have to
say something more you can't just stop
there so that's that's one point and now
the the second point is I
guess yeah so so so the the the the mark
of blanket trick is a is a very good
paper and I must admit that I personally
I am
not uh equipped uh in terms of my
knowledge of physics and uh modeling to
to to to to assess the that the
arguments they said that they use there
to to claim that the tool is
completely useless for the task that U

that it's supposed to do in the active inference framework sorry but um I think that my my and I think we would all agree here speaking for my author my co-authors I think that um what we would say about this something like this so what we would say here is well I I we we have demonstrated that Mark of blankets as such are are are instruments formal instruments that can be that do not privilege uh between any element of the of a graphical probabilistic model right so every element every variable in a probabilistic directed graphical model can be described as having its Mark of blanket or a mark of blanket is better to say a mark of blanket of that variable can be identified yeah and so if this uh if this model does not if this tool does not disambiguate between or does not privilege any of the variables in the model that means that uh when you're trying to when you're moving in the realist Direction and when you're trying to find when you're claiming that there's a boundary in the environment that will model and capture using this notion it means that you're building in some assumptions which will help you privilege certain variables over other variables in in your model and that's as long as as I said before as long as you're staying uh you know honest about that there's nothing wrong with that scientists do that all the time basically building any model requires

you to say like Hey we're going to build
in some assumptions and that's
fine and now I guess what I find
troubling and and and where the
problematic jump would be is to say okay
so if you're if you're
um if you're honest about the fact that
you're building a
representation of a
certain object in the world right and
you're honest about the assumptions so
the idealizations and the kind of you
know falsities and the simplifications
that you've built into that
model if you're honest about
that it's near impossible to make the
jump into literalism and claiming that
you know like you you have to be so in
order to be a real a good realist let's
call them has to be honest about the
assumptions that go into their model and
has to be honest about the fact that you
know all models lie in a way in some way
and in order to jump from there to to
literalism you have to become a kind of
a dishonest realist so you have to you
to abandon you know hide all the
assumptions in order to be able to do
because because otherwise it's just ER
otherwise I think at least to my mind
it's it's just this kind of epistemic
jump is impossible to make if you're
taking in all the assumptions and all
the idealizations and all the
simplifications if you're stating them
explicitly in the premises and you're
clear that they are simplifications and
idealizations then it's just basically

impossible to make this final move to
claim that the map is the territory
because you just admitted that it's a
map right so so you you kind of um yeah
I so okay so yeah I understand that in
terms of a strict literalism that the
world is always too complicated in some
sense it always udes us uh and so when
we're modeling it we always have to
build in our assumptions but it still
seems to me that there's a difference
between assumptions that lie and
assumptions that
simplify right so I can I can take a
very coar grained uh model of reality
which never
nevertheless has a faithful reliable
mapping to that reality like a map right
and and so maybe I end up having a kind
um teleonomic analysis of my map that it
serves me right like my GPS serves me it
isn't the world but it serves me now
that seems very different from a map a
mapping which lies right a map which
says that I'm in Paris when I'm in
London now that's just to one that's
totally useless and two like it's
literally wrong it's not just that it's
a Well it's not just that it's a
compression it's a compression which in
its ISO you know which in its mapping is
wrong right I'm not saying that the
mapping I'm not saying that of course a
compression is right of course there
there's not a strict isomorphic mapping
between the map and the territory
nevertheless I still think I I'm curious
about whether you could whether you

might be open to the idea let's say that
nonetheless we build in our assumptions
they're simplifying so we can't adopt a
strict literalism

nevertheless there is a way in which we
could build out of that model that we
end up having given the assumptions that
we've had and we inflate it all and we
end up with a picture of reality like an
actual true faithful picture of reality
and that follows bayesian
mechanics

um okay so well it's worth I have no
invested uh usion in this game right
like I'm I'm just asking questions so
just on the table I just think this
stuff is

interesting yeah no no this is this is
this is a perfectly fine question and I
think it's it's uh great that you're
asking it it's just that I'm trying
to uh figure out you know so so so your
question basically is that um well so so
I think we have to we have to ask
ourselves what is the end goal here
maybe this is this is what we could
start with because so literalism would
be a view that kind of like in the B has
famous short story about the Empire and
the map of the Empire literalism would
be a view where the the you know you
blow up the models you you you kind
of build that build up build up build up
build up you describe more and more
phenomena using those models and
suddenly it turns out that you've it's
the world the model it's the world
exactly so so there would be a onetoon

correspondence between the model and the world and I think that first of all that's not really feasible well it's not a model exactly it stops being a model and then the second thing is that um it's it stops having any use so to speak right uh so so it becomes useless so as as soon as the map of the Empire spans the whole empire it's it cannot perform its function as a map do you think there are people the active inference Community who think that liter you know adopt such a radical literalism no no no but but but I I'm sure that you know the literature has exploded expanded so much that I'm pretty sure pretty sure that any position you can imagine in that literature has been taken by someone so sure but uh but when it comes to the but but people do talk about you know organisms having Mark of blankets and I find this very puzzling so this is just what we were attacking this as we said earlier category mistake now the the the interesting question here the there's a second possibility here which I think you're pointing to which is that we're building we're starting with a model of the world which you know is not not an exact replica of the world it's a it's a model and then we kind of expand this model and we end up with something that is not an exact replica of the world because it shouldn't be but it's is a complete system um that corresponds in interesting ways or in relevant ways or

some truth preserving ways to how things are in the world so we kind of get this this this model that tracks things and and in a realistic way and I think that's uh I'm not close to this possibility I just don't entirely see right now how you could get there so to speak with the with the mark of blankets and active inference at this point in time so one of my worries that we don't mention in the paper I think is I don't remember now exactly but I don't think it's in the paper one of my worries is that um you can have two types of explanations right so you can have this kind of horizontal view of explanation where you're explaining you're covering a huge ground and you're bringing a lot of phenomena in under one kind of shallow umbrella where you're creating this kind of uh spreading structure that connects many many often very dist disci by using a kind of a lingua franca of of some formalism and this seems to be the project of of active inference right now so there's more and more fields that are kind of brought Under the Umbrella of free energy more and more phenomena are described as versions or as you know um of ways of doing active inference and that's very interesting it's an interesting unificatory project where you're trying to unify many different branches and many different domains of science but there's another model of explanation in science which is not about unification but is about vertical integration right so you're kind of

trying to study one phenomenon on different scales and and you're doing it in a very kind of narrow window uh narrow narrow kind of uh window of focus so you're really trying to understand how one mechanism is realized on different levels of organization so neuronal spiking right for example you can study the ion channels and then you you can study protein folding and all these kind of different aspects of how a neuron operates and you're just interested in one in one piece of the brain you're interested in how a single neuron responds to stimulate but you can study that on ever more fine grain levels of description and in the end hopefully you have this kind of vertical integration where you understand how the brain works starting from a very you know s a subcellular level onto onto the level of the whole brain or at least an area of the brain or a network of neurons uh and I'm not entirely sure that these two models of explanation are compatible so I think that active inference and the free energy principle is doing an interesting interesting work bringing in different domains of inquiry into a common framework in which they can start to to interact so suddenly you know it's it it used to be the case that people doing reinforcement learning were very far from people doing I don't know something like uh phology yeah for example yeah uh or or doing perceptual modeling and suddenly

there's a possibility that these two people these two groups of people will interact and will be you know talking to each other and that's extremely valuable however I'm not entirely sure that the free energy principle and the active inference stuff is very good at providing this kind of vertical integration precisely because as you're trying to describe everything using the same tool it might turn out that on different levels of description there are better suited more accurate ways of modeling things that don't conform to the assumptions of the free energy principle or the assumptions of the active inference framework in general whatever you want to call it so there might be kind of in different domains of inquiry kind of these vertical domains of you know where people are trying to study a particular phenomenon at a ever finer level of detail there might be domain specific or level specific ways of modeling certain certain phenomena or processes which are better suited for that purpose than the free energy formulation using Mark of blankets and then the problem becomes kind of that well should we abandon that in favor of this overarching you know structure that connects everything I think we can be a kind of pluralist about that so we can we can think of the free energy principle or the active inference framework as this uh highlevel abstract unifying layer while not uh diminishing

the work that's being done in very particular uh areas and the reason sorry just I lost track a little bit because I'm getting tired yeah soon but the reason why I'm not sure that active inference can do this vertical integration is because precisely it's trying to redescribe every phenomenon on every level of fineness of grain as doing the same thing yeah and that's actually a problem right because if you find out that a single neuron is doing the same thing that a that I don't know an ion channel is doing you might be in a you might have a problem here and I think that's that that that's maybe my my general qualm here yeah so I'm getting tired as well so so we will wrap up imminently um but there just several things that come to mind I mean one is that I think it's probably not the case at least in most recent formulations that active inference would say that the neuron is doing the same thing as a planet it's just that the neuron looks like it's doing the same thing as a planet or can be described as if it's doing the same thing as a planet and then I guess all of these concerns get deflated in some ways which is which I mean look given my again you have been in this game a lot longer than I have but even my first encounter of active influence is very much being on this looks as if line so I think we're not really going to have any problems with that and even in your commentary your response to the 35

people who got in touch uh you very much
say you know in response to inz and
Maxwell and Thomas vanz like we don't
really disagree like it's
instrumentalism deflates all of this I
guess the thing is right like you said
something really interesting in there
and I'm wondering whether it's a like
whether you meant to say it which is one
is that there may be tools which are
better at modeling right so I actually
hear this a lot from RL people where I
work at Royal Holloway which is active
influence is an as if it's very good
it's very interesting it's cool that you
can get from basic physics to uh a as if
physics of life like it's cool it's
interesting however uh we have to do
model comparison and if active inference
models uh better than RL then we'll use
it but if RL is better than active
influence we'll use that now that's
that's one thing and and like I think
everyone in the scientific world is
interested in that and and should
probably advocate for that because why
not right like it has better Downstream
uh effects when it comes to whatever
therapeutic goals but you also said
something that is that sometimes those
models those RL models uh again someone
can go back and check the tape but it
was something akin to and you can check
you can correct me uh it they may
actually break the assumptions of active
inference now that seems a different
that's true well I guess my question
here is is it a different problem

because so if I break the Assumption of a theory is it does it turn out that that theory is wrong so if for example um active inference rests on I don't know whatever it rests on like okay it rests on I mean maybe it does again I think another issue here is are we talking about the free energy principle are we talking about active inference as predictive processing active inference as P active so I've actually been thinking this in my own case uh Act of inference is a very loose term right when I write it I say it's a corollary of the free energy principle because KL says it's a COR and I know what that means but and then they say active inference is a process Theory I mean that 2017 FR and paper is active inference as a process Theory now the free energy principle is not a process Theory so active inferences is kind of this weird Middle Ground of broaching Bridging the free energy principle predict to processing and pomdp schema but you know that said I guess the question is right if I take one of these deep assumptions of active inference for example we are doing uh there's a KL Divergence so rather than doing Monte Carlo sampling we use the K Divergence if I break that do I in turn break active inference and and how deep does that rot go because right like in my estimation I've said this on this podcast Cole said it to me the free energy principle is quite shallow like it's very deep and

shallow in an interesting way it's

deflation

it doesn't really tell you anything that

exists and then you build out of it I'm

wondering whether you say Okay RL breaks

a core assumption I don't know what that

would look like but let's say it breaks

a core assumption um how deep does that

rot

go okay so what I was thinking about

specifically is that if you want to be a

real realist about Mark of

blankets you have to bring some

assumptions on board right so like you

you it's not like it all falls out of

using variational base you have to

assume some extra stuff and so in the

past one of these assumptions was

ergodicity I was gonna say ergodicity

yeah and so ergodicity is a good example

where if you have a model in some domain

at there was a time I don't know what it

is now because you know I've kind of

work on other stuff now so I have been

paying less attention to the development

in the literature but I know that people

have been moving away from the

ergodicity Assumption and the in the

inference literature but there's a

there's but ergodicity is a good example

because there was a time uh where

ergodicity was kind of baked in to what

people thought active inference and the

free energy principle are supposed to be

and then if you break that assumption so

if you don't have that assumption in

your model then it would seem that you

know you're you're you're breaking some

you're incompatible on a very important assumption with with what's happening in the what was happening at the time in the active inference and so the question would be then like does that

completely

invalidate your model or whether that completely invalidates the free energy principle and I think I would be rather reconciliatory here by saying that you know you don't expect all the models to have all the same assumptions

because models are representations of reality so what you probably want to do is like there's some phenomenon that you're trying to describe and the more complex the phenomenon like life itself the more angles you want to look at it from so you probably want to have this plurality of models and I think that's that's all fine right uh the other thing is uh and then if if you're a pluralist in this way I guess the problem you might encounter is that why would you need a single principle like the free energy principle right yes

uh in that kind of raises different questions and and again if if you're a pluralist if you're a kind of a perspectival list where you think that there are multiple perspectives to look at single phenomenon that are can be all equally informative depending on your explanatory interests or pragmatic interests then I would say Well it starts to become it starts to be difficult to see why you would want to claim that there's this overarching

principle yeah unless you're looking at this Principle as just a means as I said before of building this super structure right which is just you know as if doing free energy minimization story which is supposed to create these Bridges between very distant disciplines or domains bringing them into together into into conversation which is pragmatically very valuable let's agree on that but it's it's it might not be the be all end all of you know science no no I mean I mean yeah there's an interesting kind of meta sociological point about why people want to Gran unifying theory of anything I mean it's it's definitely satisfying um in some primordial sense but beyond that it may be wise to rest among um plurality as you say and and there's also the you know the argument from the pessimistic meta induction which is that uh every Theory up till now so we have certain set of scientific theories we think are uh best right now but every theory that preceded them in each of the fields has been discarded as false or as you know missing some important aspects so why should we assume that our current theories are true if history of science is is you know layered with with with husks of abandoned theories and one more thing that I just wanted to mention you earlier said about the incompatibility of different theories in science and I immediately thought about the incompatibility of quantum mechanics and

and Newtonian mechanics and what's interesting about that case is that even though we don't really know how to make the two compatible uh we haven't abandoned either right so there there's like there's a possibility that you will have on different levels of description of reality you will have different models with different assumptions describing different phenomena and that's actually okay maybe that's just how it is yeah I I'd like to get a philosopher philosopher science on here because I think um this isn't really well this isn't my era of expertise but I I think there are slight differences between the models that employed by the free energy principle and the models that Newton employed and I think one of them is that when I write down uh $f = ma$ right the of course I make the Assumption of you know and I do that school and I they say okay what's the acceleration of the car and it's got this got this Mass I then make the Assumption of uh win resistance and um the lack of win resistance Tire lack of friction and everything like but it seems to me that in principle Newton could have recognize those assumptions and built them into his model and he gets a closer approximation of the truth the question therefore and we'll leave it here because we been going for three hours and a bit is whether the free energy principle is like Newtonian mechanics such that it can act as the base a true

ontological base because I think if you ask Newton he would say literally $f = ma$ right like it's not just I say it and it looks as if things are doing $f = ma$ it's like if I know like it's not just if I know M and I know a going to know F it's like that's a law of the universe inscribed in the universe now I'm not a philosopher of science I'm also not a physicist so I might be wrong here but it seems to me that that therefore acts as a base on which you can build in friction you can build in air resistance and so on and you get a closer approximation on the truth and that doesn't make his theory false right because it idealizes it just makes it a base right and it just means it's pragmatically useful but it doesn't make it false the real question here therefore and I'm not going to answer it today because I'm out of caffeine and I'm out of energy is whether the free and liy principle is that base as well and I guess uh time will tell on that front but you know I mean ultimately epistemic humility I do think is the way right I mean I think I think if we can talk about one recurrent thing that's been very useful in the history of science is epistemic humility plurality listening to the other side and uh so that's why you know I very much appreciate you coming on the show um because well it's not a lion's den I mean I've had viento on I've had all sorts I've had people who don't I think most people are partial to active

inference but it's not really a strict
active inference fan club um but no I
mean I really do appreciate the openness
and the expertise
and the dantian curiosity now it's um
it's been amazing I mean three hours and
five minutes have flown by thanks yeah
it's sorry that I wore you out uh so
much you should you should be
apologizing if you didn't wear me out um
uh yeah you just to close up you asked
me at some point who wrote the maths in
the mark of blanket paper so it was
first and foremost Manuel baltieri
because he has background in machine
learning so he knows a lot of this stuff
and then I was helping him I think that
was the that would be the correct way to
nice okay yeah always I'm always I'm
always super impressed I mean it's just
it's kind of a project that I'm trying
to undertake myself as well orbe it
having that and then simultaneously
battling the kind of fears like no this
is just an instrumental tool get back to
Consciousness get back to Consciousness
like that's for not like you know at
least that exists and so on and there's
an
interesting uh sort of pulling and you
know throwing in different directions
and I guess active inference does that
it Bridges as you say these multiple
domains but I'm very glad personally
people might disagree with me I'm very
glad that philosophers have brought to
bear their expertise um I think as you
know in the modern world we probably are

a bit disenchanted by philosophy but actually I think with the prominence of active inference it's kind of shown its strength and power to say hold on a second think about your presuppositions think about your axioms and I'm personally a really big fan that of the fact that phenomenology and Consciousness has also kind of returned I mean it's not that you know it returned in 1995 or you know you can go Thomas Nagel but like this feels like a real Renaissance of this question um so you know Long Live philosophy I think is the is the final note but yeah Chris amazing I I think this much well I think this is the longest podcast um no no no no well Cole went no way Cole went for about three hours Julian knee went for about three hours so you're you're in good company okay thank you really I this was really really fun I really enjoyed it so thank you so much uh all right thank you so much for your time again sorry for wearing you out sorry if I was maybe a little bit incoherent at times which sometimes happens when you're going for so long uh please send me the draft of your paper you have my email so I would love to read that that's very kind and also Al uh before you leave and before we end the show please let people know where they can find you what you've got coming up you'll work with ax or like anything that you want to say in terms of self right uh I know this horrible

everyone hates this bit but no no it's
yeah I I get it that's how it works um
so my website is Chris k r Ys
dolega

doxyz uh that's where you can kind of
track my Publications and stuff and if
you look for chriso Lega spelled in the
same way on Twitter or blue sky and
Mastadon I think that will also pop
up um and uh yeah together with Axel
we're organizing a symposium at this
year's assc in Tokyo oh nice so that's
something that uh if you're in Tokyo at
the time and visiting for the assc you
can come to it's on unconscious affect
and finally I will be also giving a talk
which will be online uh in something
called uh the Tokyo analytic philosophy
Forum uh let me just check if I remember
the date

uh I will make sure all of this is in
the uh Tokyo forum for analytic
philosophy it's it's uh series of talks
that's hosted at the Tokyo University
and I will be speaking on on the 19th of
July fantas actually about something
that we touched on briefly here so about
effective modes of presentation and
perceptual experience so yeah if you're
interested in that you can join that
will be that will be online will be
going in this will be going in the
description assc will be going in the
description I'll put the website in that
as well okay cool cool we've done three
hours and 10 minutes okay amazing thank
you so much thanks a lot have a good
evening and take care